

Regence

Medical Policy Manual

Topic: Stereotactic Radiosurgery and Stereotactic Body Radiation Therapy

Date of Origin: January 1996

Section: Surgery

Last Reviewed Date: April 2016

Policy No: 16

Effective Date: May 1, 2016

IMPORTANT REMINDER

Medical Policies are developed to provide guidance for members and providers regarding coverage in accordance with contract terms. Benefit determinations are based in all cases on the applicable contract language. To the extent there may be any conflict between the Medical Policy and contract language, the contract language takes precedence.

PLEASE NOTE: Contracts exclude from coverage, among other things, services or procedures that are considered investigational or cosmetic. Providers may bill members for services or procedures that are considered investigational or cosmetic. Providers are encouraged to inform members before rendering such services that the members are likely to be financially responsible for the cost of these services.

DESCRIPTION

Stereotactic radiosurgery (SRS) and stereotactic body radiotherapy (SBRT) or stereotactic ablative radiotherapy (SABR) are radiotherapy techniques that use highly focused radiation beams to treat both neoplastic and non-neoplastic conditions, in contrast to traditional external radiation beam therapy, which involves the use of relatively broad fields of radiation over a number of sessions that may occur over weeks to months. The term SBRT will be used to describe treatment also referred to as SABR. SRS and SBRT rely on three-dimensional imaging to localize the therapy target. SRS and SBRT have been used for a range of malignant and non-malignant conditions. Because they are more targeted than traditional external radiation therapy, SRS and SBRT are often used for treatment at sites that are difficult to reach via surgery, located close to other vital structures, or subject to movement within the body.

Both SRS and SBRT may be completed with one session (single-fraction) or may require additional sessions over a course of days, referred to as fractionated stereotactic radiotherapy. The fractionation used for SRS and SBRT is referred to as “hypofractionated” because it is less than that used for conventional external beam radiotherapy.” Fractionation of stereotactic radiotherapy aims to optimize the therapeutic ratio; that is the ratio between tumor control and late effects on normal tissues. The main advantage of fractionation is that it allows higher total doses to be delivered to the tumor because of

increased tolerance of the surrounding healthy tissues to each individual, fractionated dose. In addition, some lesions such as large arteriovenous malformations may require more than one procedure to complete the obliteration process.

SRS and SBRT can be administered by several types of devices that are distinguished by their source of radiation, including particle beams (e.g., proton), gamma radiation from cobalt-60 sources, or high-energy photons from linear accelerator (LINAC) systems. The Gamma Knife and linear accelerator systems (including the Cyberknife®) are similar in concept; both use multiple photon radiation beams that intersect at a stereotactically determined target, thus permitting higher doses of radiation delivery with sparing of surrounding normal tissues. The differences between the two relate to how the energy is produced (i.e., through decaying cobalt-60 in the gamma knife devices, or from x-rays in the linear accelerator system) and the number of energy sources used (i.e., multiple energy sources in the gamma knife versus one in the linear accelerator system).

Stereotactic Radiosurgery (SRS)

The most common applications of SRS include treatment of intracranial malignancies. In order to localize the treatment target and to achieve precise delivery of radiation, SRS relies on stereotactic guidance (using cerebral angiography, computerized tomography, and/or magnetic resonance imaging) and the use of a positioning frame to restrict head movement.

Stereotactic Body Radiation Therapy (SBRT)

SBRT refers to the use of SRS-like technology for extracranial sites. Radiation may be delivered in a single high dose or a small number of fractionated treatments. SBRT is made possible by the recent availability of repositioning devices that can be used to restrict body movement. SBRT may also be referred to as stereotactic ablative radiotherapy (SABR).

Image-Guided Radiosurgery or Radiotherapy

Image-guided radiosurgery or radiotherapy is a relatively new development collectively describing units with real-time image guidance systems. Examples include the Cyberknife® device, BrainLAB Novalis®, TomoTherapy®, and LINAC with computerized tomography (CT).

Regulatory Status

Several devices that use cobalt 60 radiation (gamma ray devices) for SRS have been cleared for marketing by the U.S. Food and Drug Administration (FDA) through the 510(k) process. The most commonly used gamma ray device is the GammaKnife (Elekta; approved May 1999). Gamma ray emitting devices that use cobalt 60 degradation are also regulated through the U.S. Nuclear Regulatory Commission.

A number of LINAC movable platforms that generate high-energy photons have been cleared for marketing by the FDA through the 510(k) premarket notification process including the Novalis Tx® (Novalis, Westchester, IL); the TrueBeam STx (Varian Medical Systems, Palo Alto, CA, approved December 2012); and the CyberKnife® System (Accuray, Inc.; approved December 1998). LINAC-based devices may be used for intracranial and extracranial lesions.

Note: Particle radiation can also be used without stereotactic guidance. In this setting, the use of particles is referred to as proton, helium, or neutron radiation *therapy*. Proton or helium ion radiation therapies (RT), intraocular RT for age-related macular degeneration, and electromagnetic navigation bronchoscopy for placement of fiducial markers are considered in separate medical policies. See cross-reference section below.

MEDICAL POLICY CRITERIA

- I. Stereotactic radiosurgery (SRS) including fractionated stereotactic radiotherapy may be considered **medically necessary** for the following indications:
 - A. Acoustic neuromas (Vestibular Schwannomas)
 - B. Arteriovenous malformations
 - C. Chordomas and chondrosarcomas, of the skull base
 - D. Craniopharyngiomas
 - E. Hemangioblastoma
 - F. Hemangiopericytoma as primary treatment or salvage treatment of recurrence
 - G. Glomus jugulare and Glomus tympanicum tumors
 - H. Meningiomas that are nonresectable, residual, or recurrent
 - I. Metastasis to CNS (solitary or multiple) in patients when both of the following are met:
 - 1. Karnofsky performance score ≥ 70 or an ECOG score ≤ 2 (See Policy Guidelines); and
 - 2. Life expectancy > 6 months.
 - J. Pituitary adenomas
 - K. Primary neoplasms of the CNS, including but not limited to low grade gliomas and high-grade gliomas
 - L. Trigeminal neuralgia (tic douloureux) refractory to medical management
 - M. Uveal melanoma
- II. Stereotactic body radiation therapy (SBRT) (i.e., stereotactic ablative radiotherapy [SABR]) including fractionated stereotactic radiotherapy may be considered **medically necessary** for the following indications:
 - A. Hepatocellular carcinoma when all of the following criteria are met:
 - 1. ≤ 5 hepatic lesions; and

2. Lesion(s) is/are ≤ 6 cm diameter; and
 3. Karnofsky performance score ≥ 70 or an ECOG score ≤ 2 (See Policy Guidelines); and
 4. Life expectancy > 6 months.
- B. Primary or metastatic hepatic tumor as palliative or curative treatment when all of the following are met:
1. Absence or minimal extra hepatic disease; and
 2. Karnofsky performance score ≥ 70 or an ECOG score ≤ 2 (See Policy Guidelines); and
 3. Life expectancy > 6 months.
- C. Lung metastases when all of the following criteria are met:
1. ≤ 5 metastatic lung lesions; and
 2. Karnofsky performance score ≥ 70 or an ECOG score ≤ 2 (See Policy Guidelines); and
 3. Life expectancy > 6 months.
- D. Non-small cell lung cancer (NSCLC), primary (node negative, tumor stage T1a, T1b, T2a, T2b)
- E. Osteosarcoma, metastatic when all of the following criteria are met:
1. ≤ 5 metastatic lesions; and
 2. Karnofsky performance score ≥ 70 or an ECOG score ≤ 2 (See Policy Guidelines); and
 3. Life expectancy > 6 months.
- F. Spinal or paraspinal tumors (primary or metastatic) including initial treatment or salvage treatment of local recurrence after previous irradiation when there is clinical documentation of no high grade epidural compression of the spinal cord

III. Stereotactic radiosurgery and stereotactic body radiation therapy are considered **investigational** for all other indications including but not limited to:

- A. Choroidal neovascularization (CNV)
- B. Chronic pain

- C. Epilepsy
- D. Functional disorders other than trigeminal neuralgia
- E. Primary tumors of the following primary sites or metastatic tumors to the following sites:
 - 1. Cervix
 - 2. Endometrium
 - 3. Esophagus
 - 4. Hemangiomas
 - 5. Kidney
 - 6. Large bowel
 - 7. Ovaries
 - 8. Pancreas
 - 9. Prostate
 - 10. Rectum
 - 11. Small bowel
- F. Refractory symptoms of essential tremor or Parkinson's disease
- G. Seizures

POLICY GUIDELINES

Performance Status Measurement

Performance status is frequently used in oncology practice as a variable in determining prognosis and management strategies. Either the Karnofsky Performance Status (KPS) or the Eastern Cooperative Oncology Group (ECOG) Performance Status scoring systems may be used.

<i>Karnofsky Performance Status</i>			
100	Normal, without symptoms	50	Requires considerable assistance and frequent medical care
90	Able to carry on normal activity; minor signs or symptoms of disease	40	Disabled; requires special care and assistance
80	Normal activity with effort; some signs or symptoms of disease	30	Severely disabled; hospitalization is indicated

70	Cares for self; unable to carry on normal activity or do active work	20	Very sick; active support treatment is necessary
60	Requires occasional assistance; able to care for most personal needs	10	Moribund; fatal processes progressing rapidly
<i>ECOG Performance Status:</i>			
0	Fully active, able to carry on all pre-disease performance without restriction		
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work		
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours.		
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.		
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.		

Fractionation

Fractionated stereotactic radiotherapy refers to when SRS or SBRT are performed in 1-10 treatments focused upon a specific site. SRS is commonly delivered in 1-5 fractions and SBRT or SABR is commonly delivered in 1-5 fractions but may be delivered in as many as 10 fractions.

Dose Constraint References

Radiation Therapy Oncology Group (RTOG) Radiation Dose Constraints

Available from: https://en.wikibooks.org/wiki/Radiation_Oncology/Toxicity/RTOG

Stereotactic Radiosurgery (single fraction)

Quantitative Analyses of Normal Tissue Effects in the Clinic (QUANTEC)

Available from: https://en.wikibooks.org/wiki/Radiation_Oncology/Toxicity/QUANTEC

SCIENTIFIC EVIDENCE

The selection of variables used in the delivery of SRS and SBRT is complex and individualized, requiring selection of the device, radiation dose, and the size and shape of treatment margins. All of these variables depend on the location, shape, and radiosensitivity of the target tissue and the function and radiosensitivity of the surrounding tissue. Trials that allow direct comparison of all of the possible variables involved in selecting specific SRS and SBRT methods do not broadly exist making it difficult to draw comparative effectiveness conclusions. Further, for many rare conditions, large comparative studies are unlikely.

Please note that the evidence review below does not compare specific radiation planning and delivery techniques.

Arteriovenous Malformations

In 2014, Mohr et al reported results of the ARUBA trial, a randomized, multicenter trial to compare medical therapy with medical therapy with interventional therapy (including any neurosurgical, endovascular, or stereotactic radiotherapy procedure) in patients with unruptured arteriovenous malformations (AVM).^[1] Two hundred twenty-six patients were enrolled and randomized, 116 to interventional therapy and 110 to medical management. Among those randomized to interventional therapy, 91 received interventional therapy, 5 with neurosurgery alone, 30 with embolization alone, 31 with radiotherapy alone, 12 with embolization and neurosurgery, 15 with embolization and radiotherapy, and 1 with all 3. The trial was stopped early by its data and safety monitoring board after interim analysis demonstrated superiority of medical management, after outcomes were available for 223 patients with mean follow-up time of 33.3 months. The risk of death or stroke was lower in the medical management group than in the interventional therapy group (hazard ratio [HR], 0.27; 95% confidence interval [CI], 0.14 to 0.54). Patients will continue to be followed to determine whether differences in outcomes persist. Although a high proportion of patients randomized to interventional therapy (40.5%) received at least some radiotherapy, outcomes are not reported by therapy type, making it difficult to assess the comparative effectiveness of SRS in AVM treatment.

Paul et al conducted a retrospective cohort study that included 697 SRS treatments in 662 patients treated with SRS for brain AVMs at a single institution.^[2] The obliteration rate after a single or multiple SRS procedures was 69.3% and 75%, respectively. The obliteration rates were significantly associated with AVMs that were compact (odds ratio [OR], 3.16; 95% CI, 1.92 to 5.22), with undilated feeders (OR=0.36; 95% CI, 0.23 to 0.57), with smaller volume (OR=0.95; 95% CI, 0.92 to 0.99) and that were treated with higher marginal dose (OR=1.16, 95% CI, 1.06 to 1.27).

Bowden et al reported outcomes from a retrospective cohort study of patients with cerebellar AVM treated with SRS at a single institution.^[3] Sixty-four patients were included, 73% of whom had presented with intracranial hemorrhage and 19% of whom had undergone prior embolization. Total obliteration was achieved at 3, 4, and 5 to 10 years in 52%, 69%, and 75%, respectively, of subjects. Obliteration was more likely in smaller AVMs but less likely in patients who had undergone prior embolization. Symptomatic adverse radiation events, defined by magnetic resonance imaging (MRI) changes and new neurologic deficits in the absence of hemorrhage, occurred in 3 patients.

Fokas et al reported long-term follow-up of a cohort of patients who underwent SRS for cerebral AVMs at a single institution.^[4] One hundred sixty-four patients were identified, with a median follow-up of 93 months (range, 12-140 months). Thirty-nine percent of subjects had experienced a prior intracranial hemorrhage, and 43.3% and 8.0%, respectively, had undergone prior embolization or neurosurgical procedures. Complete obliteration was seen in 61% of patients at a median time of 29 months. Complete obliteration was achieved at 3 and 5 years in 61% and 88%, respectively. In multivariable models, higher radiation dosage and smaller target volumes were associated with higher rates of complete obliteration. The annual bleeding risk was 1.3% per year during follow-up.

Matsuo et al reported outcomes from a cohort of 51 patients with intracranial AVMs treated with SRS at a single institution.^[5] Rates of obliteration after a single SRS at 3, 5, 10, and 15 years were 46.9%, 54%, 64%, and 68%, respectively; rates of obliteration after multiple SRS sessions at 3, 5, 10, and 15 years were 46.9%, 61.3%, 74.2%, and 90.3%, respectively. The adverse radiation events occurred in 9 cases (17.6%), with 4 cases (3 symptomatic cysts and 1 intracranial hemorrhage) not occurring until 10 years after the SRS treatment.

Potts et al summarized outcomes for 80 children treated with SRS for intracranial AVMs, most of whom (56%) had intracranial hemorrhage at the time of presentation.^[6] Among the 47% of subjects with available angiograms 3 years after treatment, AVM obliteration occurred in 52% of patients treated with higher-dose SRS (18-20 Gy) and in 16% treated with lower-dose SRS (<18 Gy).

Kano et al reported a study to define long-term outcomes and risks of AVM management using 2 or more stages of SRS for symptomatic large-volume lesions unsuitable for surgery.^[7] Forty-seven patients with such AVMs underwent volume-staged SRS. Eighteen patients (38%) had had a prior hemorrhage and 21 patients (45%) had undergone prior embolization. In 17 patients, AVM obliteration was confirmed after 2 to 4 SRS procedures at a median follow-up of 87 months (range, 0.4-209 months). Five patients had near-total obliteration (volume reduction >75% but residual AVM). The actuarial rates of total obliteration after 2-stage SRS were 7%, 20%, 28%, and 36% at 3, 4, 5, and 10 years, respectively. The 5-year total obliteration rate after the initial staged volumetric SRS was 62% (p=0.001). Sixteen patients underwent additional SRS at a median interval of 61 months (range, 33-113 months) after the initial 2-stage SRS. The overall rates of total obliteration after staged and repeat SRS were 18%, 45%, and 56% at 5, 7, and 10 years, respectively. Ten patients sustained hemorrhage after staged SRS, and 5 of these patients died. Three of 16 patients who underwent repeat SRS sustained hemorrhage after the procedure and died. Based on Kaplan-Meier analysis (excluding the second hemorrhage in the patient who had 2 hemorrhages), the cumulative rates of AVM hemorrhage after SRS were 4.3%, 8.6%, 13.5%, and 36.0% at 1, 2, 5, and 10 years, respectively, corresponding to annual hemorrhage risks of 4.3%, 2.3%, and 5.6% for years 0 to 1, 1 to 5, and 5 to 10 after SRS. Multiple hemorrhages before SRS correlated with a significantly higher risk of hemorrhage after SRS. Symptomatic adverse radiation effects were detected in 13% of patients. The authors concluded that volume-staged SRS for large AVMs unsuitable for surgery has potential benefit, but often requires more than 2 procedures to complete the obliteration process and that in the future, prospective volume-staged SRS followed by embolization (to reduce flow, obliterate fistulas, and occlude associated aneurysms) may improve obliteration results and further reduce the risk of hemorrhage after SRS.

Section Summary

The evidence related to the use of SRS for AVM consists primarily of noncomparative cohort studies, which demonstrate relatively high rates of complete obliteration of AVM after SRS, in the range of approximately 40% in some studies to greater than 70% in others. Isolating the effect of the SRS therapy in and of itself can be challenging, as many patients are treated with more than one therapy, including endovascular treatments and surgery. Recently, an RCT that compared medical therapy with a variety of interventions in the treatment for AVM showed no significant improvement in outcomes with interventional therapy. However, given that the interventional therapies included a variety of therapies, it is difficult to assess whether one particular component of the intervention has benefit or lacks benefit. Longer-term follow up will be forthcoming from this study.

Trigeminal Neuralgia

A 2011 review article summarizes the literature on the use of SRS for trigeminal neuralgia.^[8] Most patients with typical facial pain will achieve relief following radiosurgical treatment.

Dhople reports long-term outcomes of SRS for classical trigeminal neuralgia in 112 patients treated between 1996 and 2001.^[9] Of these, 67% had no prior invasive operations for trigeminal neuralgia (TM) prior to SRS, 13% had 1, 4% had 2, and 16% had 3 or more. The right side was affected in 56% of cases, predominantly involving V2 (26%), V3 (24%), or a combination of both (18%) branches. The

median age at diagnosis was 56 years, and median age at SRS was 64 years. The median prescription dose of 75 Gy (range, 70-80 Gy) was delivered to the involved trigeminal nerve root entry zone. The authors assessed the degree of pain before and after SRS by using the Barrow Neurological Institute (BNI) pain scale. In total, 102 patients took the survey at least once, for a response rate of 91%. Although not found to alter the conclusions of this study, 7 cases of atypical TN were found, and these patients were removed, for a total of 95 cases herein analyzed. The median follow-up was 5.6 years (range, 13-115 months). Before gamma knife surgery (GKS), 88% of patients categorized their pain as BNI IV or V (inadequate control or severe pain on medication), whereas the remainder described their pain as BNI III (some pain, but controlled on medication). After GKS, 64% reported a BNI score of I (no pain, no medications), 5% had BNI II (no pain, still on medication), 12% had BNI III, and 19% reported a BNI score of IV or V. The median time to response was 2 weeks (range, 0-12 weeks), and the median response duration was 32 months (range, 0-112 months). Eighty-one percent reported initial pain relief, and actuarial rates of freedom from treatment failure at 1, 3, 5, and 7 years were 60%, 41%, 34%, and 22%, respectively. Response duration was significantly better for those who had no prior invasive treatment versus those in whom a previous surgical intervention had failed (32 vs 21 months, $p<0.02$). New facial numbness was reported in 6% of cases.

Section Summary

Case series identify improvements in pain related to trigeminal neuralgia after treatment with SRS. Comparative studies that evaluate the use of SRS compared with alternative treatments for trigeminal neuralgia are lacking.

Epilepsy

A 1998 TEC Assessment^[10] cited 2 studies of 11 and 9 patients, respectively, in which radiosurgery was used to treat epilepsy. The subsequent literature search revealed 3 small studies on the use of radiosurgery for medically refractory epilepsy. Regis et al^[11] selected 25 patients with mesial temporal lobe epilepsy, 16 of whom provided minimum 2-year follow-up. Seizure-free status was achieved in 13 patients, 2 patients were improved, and 3 patients had radiosurgery-related visual field defects. A study by Schrottner et al^[12] included 26 patients with tumor-related epilepsy, associated mainly with low-grade astrocytomas. Mean follow-up among 24 available patients was 2.25 years. Tumor location varied across patients. Seizures were simple partial in 6 (3 with generalization) and complex partial in 18 (5 with generalization, 1 gelastic). Seizures were eliminated or nearly so in 13 patients. Little improvement was observed in 4 patients and none in 7. Whang and Kwon^[13] performed radiosurgery in 31 patients with epilepsy associated with nonprogressive lesions. A minimum of 1-year follow-up was available in 23 patients, 12 of whom were seizure-free (and 3 of whom had antiseizure medications discontinued), 2 had seizures reduced in frequency, and 9 experienced no change. While the Regis series selected a fairly homogeneous clinical sample, the other 2 studies were heterogeneous. No confirmatory evidence is available on mesial temporal lobe epilepsy. The available evidence from patients with epileptic lesions of various sizes and locations is insufficient to show what factors are associated with favorable outcome.

In the most recent literature review (2014), no new comparative studies evaluating SRS for the treatment of epilepsy were identified.

Section Summary

The currently-available research related to the use of SRS for epilepsy treatment is preliminary. There is inadequate information to determine the risk: benefit ratio of SRS compared with other therapies for epilepsy treatment.

Tremor

SRS has been used for the treatment of tremor through stereotactic radiofrequency thalamotomy. In 2008, Kondziolka et al reported outcomes for 31 patients who underwent SRS thalamotomy for disabling essential tremor.^[14] Among 26 patients with follow-up data available, score on the Fahn-Tolosa-Marin tremor score improved compared with baseline from 3.7 (pre-SRS) to 1.7 (post-SRS; $p<0.000$) and score on the Fahn-Tolosa-Marin handwriting score improved compared with baseline from 2.8 (pre-SRS) to 1.7 (post-SRS; $p<0.000$). One patient developed transient mild right hemiparesis and dysphagia and 1 patient developed mild right hemiparesis and speech impairment.

Kooshkabadi et al reported outcomes for 86 patients with tremor treated over a 15-year period, including 48 with essential tremor, 27 with Parkinson disease, and 11 with multiple sclerosis.^[15] Fahn-Tolosa-Marin tremor scores were used to compare symptoms pre- and postprocedure: the mean tremor score improved from 3.28 (pre-SRS) to 1.81 (post-SRS; $p<0.000$), the mean handwriting score improved from 2.78 (pre-SRS) to 1.62 (post-SRS; $p<0.000$), and the mean drinking score improved from 3.14 (pre-SRS) to 1.8 (post-SRS, $p<0.000$). Complications included temporary hemiparesis in 2 patients, dysphagia in 1 patient, and sustained facial sensory loss in 1 patient.

Lim et al reported outcomes for a small cohort of 18 patients who underwent SRS treatment for essential tremor.^[16] For the 14 patients with videotaped evaluations allowing blinded evaluation of tremor severity and at least 6 months of follow-up ($N=11$ with essential tremor and $N=3$ with Parkinson disease), Fahn-Tolosa-Marin Tremor Rating Scale activities of daily living scores improved significantly after SRS (mean change score 2.7 points; $p=0.03$). However, there was no significant improvement in other Fahn-Tolosa-Marin Tremor Rating Scale items ($p=0.53$ for resting tremor, $p=0.24$ for postural tremor, $p=0.62$ for action tremor, $p=0.40$ for drawing, $p>0.99$ for pouring water, $p=0.89$ for head tremor). Mild neurologic complications occurred in 2 patients (lip and finger numbness), and severe neurologic complications occurred in 1 patient (edema surrounding thalamic lesion with subsequent hemorrhage at the lesion site, with speech difficulty and hemiparesis.)

Ohye et al conducted a prospective study of SRS for tremor that included 72 patients, 59 with Parkinson disease and 13 with essential tremor.^[17] Among 52 patients who had follow-up at 24 months, tremor scores measured using the unified Parkinson's Disease Rating Scale ($p<0.001$; approximate score decrease extrapolated from graph from 1.5 at baseline to 0.75 at 24-month follow-up).

Young et al reported outcomes for a cohort of 158 patients with tremor who underwent SRS, including 102 patients with Parkinson disease, 52 with essential tremor, and 4 with tremor due to other conditions.^[18] Among patients with a parkinsonian tremor, at latest follow-up (mean, 47 months), blinded assessments on unified Parkinson's Disease Rating Scale demonstrated improvements in several specific items, including overall tremor (from 3.3 pretreatment to 1.2 at last follow-up; $p<0.05$) and action tremor (from 2.3 pretreatment to 1.3 at last follow-up; $p<0.05$). Among patients with Essential tremor, blinded assessments were conducted using the Fahn-Tolosa-Marin Tremor Rating Scale. At 1-year of follow-up, 92.1% of patients with essential tremor were completely or nearly tremor-free. Improvements were reported in components of the Fahn-Tolosa-Marin Tremor Rating Scale, but statistical comparisons are not presented. Three patients developed new neurologic symptoms attributed to the SRS.

Section Summary

The evidence related to the use of SRS for tremor consists of uncontrolled cohort studies, many of which report outcomes from the treatment of tremor of varying etiologies. Most studies report improvements in standardized tremor scores, although few studies used a blinded evaluation of tremor score, allowing for bias in assessment. No studies that compared SRS with alternative methods of treatment or a control group were identified. Limited long-term follow-up is available, making the long-term risk: benefit ratio of an invasive therapy uncertain.

Chronic Pain

The TEC Assessment from 1998 identified 2 reports, with 2 and 47 patients, respectively, who underwent radiosurgical thalamotomy for chronic pain. No new studies were identified in a search of recent literature. Thus, the conclusions of the 1998 TEC Assessment have not changed.

Central Nervous System Neoplasms

Acoustic Neuromas

SRS is widely used for the treatment of acoustic neuromas (vestibular schwannomas). Case series report generally high rates of local control (LC). For example, Badahshi et al reported a 3-year local tumor control rate of 88.9% in a study of 250 patients with vestibular schwannoma who underwent SRS or fractionated SRS.^[19] Williams et al reported rates of tumor progression-free survival (PFS) for patients with large vestibular schwannomas treated with SRS of 95.2% and 81.8% at 3 and 5 years, respectively.^[20] For patients with small vestibular schwannomas treated with SRS, tumor PFS was 97% and 90% at 3 and 5 years, respectively. In a retrospective case series of 93 patients with vestibular schwannomas treated with SRS, 83 of whom had long-term follow-up, Woolf et al reported an overall control rate of 92% at a median follow-up of 5.7 years. A small study from 2006 that compared microsurgical resection (N=36) with SRS (N=46) for the management of small (<3 cm) vestibular schwannomas showed better hearing preservation at last follow-up in the SRS group ($p<0.01$) and no difference in tumor control between the groups (100% vs 96%, $p=0.50$).^[21]

In the treatment of acoustic neuromas, the most significant adverse effect is loss of function of the facial and auditory nerve. For example, in a single-institution study, Meijer et al reported on the outcomes of single fraction versus fractionated linear accelerator (LINAC)-based SRS in 129 patients with acoustic neuromas.^[22] Among these patients, 49 were edentate and thus could not be fitted with a relocatable head frame that relies on dental impressions. This group was treated with a single fraction, while the remaining 80 patients were treated with a fractionated schedule. With an average follow-up of 33 months, there was no difference in outcome in terms of local tumor control, facial nerve preservation, and hearing preservation. Chung et al reported on the results of a single-institution case series of 72 patients with acoustic neuromas, 45 of whom received single-fraction therapy and 27 who received fractionated therapy.^[23] Patients receiving single-fraction treatment were functionally deaf, while those receiving fractionated therapy had useful hearing in the affected ear. After a median follow-up of 26 months, there was no tumor recurrence in either group. Chang et al reported that 74% of 61 patients with acoustic neuromas treated with CyberKnife using staged treatment had serviceable hearing maintained during at least 36 months of follow-up.^[24]

Section Summary

The evidence related to the use of SRS for acoustic neuroma (vestibular schwannoma) consists primarily of case series and cohort studies, which report high rates of freedom from tumor progression. Given that vestibular schwannoma is a slow-growing tumor with symptoms most often related to local

compression, demonstration of slowing of progression is a reasonable outcome. A single comparative study was identified that demonstrated comparable tumor control outcomes between SRS and surgical therapy for small vestibular schwannomas.

Craniopharyngioma

Hashizume et al evaluated the results of the use of SRS in 10 patients with craniopharyngioma adjacent to optic pathways.^[25] Ten patients (6 men, 4 women) with craniopharyngioma and median age of 56.5 years (range, 10-74 years) were treated from 2006 through 2009. Median volume of tumor was 7.9 mL (range, 1.1-21 mL). A total dose of 30 to 39 Gy in 10 to 15 fractions (median, 33 Gy) was delivered to the target. Ten patients were followed up for 9-36 months (median, 25.5 months). The response rate was 80% (8/10), and control rate was 100%. Improvement of neurologic symptoms was observed in 5 patients. No serious complications due to SRS were found.

Hasegawa et al determined the limiting dose to the optic apparatus in single-fraction irradiation in patients with craniopharyngioma treated with gamma knife radiosurgery.^[26] One hundred patients with 109 craniopharyngiomas treated with radiosurgery were evaluated with a median follow-up period of 68 months. Tumor volume varied from 0.1 to 36.0 cm (median, 3.3 cm). The actuarial 5- and 10-year overall rates of survival of tumor progression after radiosurgery were 93% and 88%, respectively. The actuarial 5- and 10-year PFS rates were 62% and 52%, respectively.

Among 94 patients in whom visual function was evaluable, only 3 patients developed radiation-induced optic neuropathy, indicating an overall Kaplan-Meier radiation-induced optic neuropathy rate of 5%.

Combs et al evaluated the long-term outcome in patients with craniopharyngiomas treated with fractionated stereotactic radiotherapy.^[27] A total of 40 patients with craniopharyngiomas were treated between 1989 and 2006. Most patients were treated for tumor progression after surgery. A median target dose of 52.2 grays (Gy) (range, 50.4-56 Gy) was applied in a median conventional fractionation of 5×1.8 Gy per week. Follow-up examinations included thorough clinical assessment, as well as contrast-enhanced MRI scans. After a median follow-up of 98 months (range, 3-326 months), LC was 100% at both 5 years and 10 years. Overall survival (OS) rates at 5 years and 10 years were 97% and 89%, respectively. A complete response was observed in 4 patients and partial responses were noted in 25 patients. Eleven patients presented with stable disease during follow-up. Acute toxicity was mild in all patients. Long-term toxicity included enlargement of cysts requiring drainage 3 months after fractionated stereotactic radiotherapy (FSRT). No visual impairment, radionecrosis, or development of secondary malignancies was observed. The authors concluded that long-term outcome of fractionated radiosurgery for craniopharyngiomas is excellent with regard to LC, as well as treatment-related side effects.

Section Summary

The evidence related to the use of SRS for craniopharyngioma consists primarily of case series and cohort studies, which report high rates OS. There is a lack of comparative studies evaluating the treatment of pituitary adenomas with SRS versus surgery or traditional radiotherapy.

Facial Nerve Schwannomas

Sheehan published a multicenter case series study that evaluated 42 patients with facial nerve schwannomas undergoing SRS.^[28] Prior resection was performed in 36% of cases. At a median follow-

up of 28 months, tumor control was achieved in 90% of patients. The study authors reported that most patients treated with SRS had neurological preservation. Smaller tumors treated with SRS had better outcomes for nerve function.

Section Summary

The evidence related to the use of SRS for facial nerve schwannomas consists of case series, which report high rates of tumor control and nerve preservation. However, there remains a lack of comparative studies that evaluate long term outcomes including overall survival.

Glomus Jugulare Tumors

Ivan et al conducted a meta-analysis of tumor control rates and treatment-related mortality for patients with glomus jugulare tumors.^[29] In this study, the authors assessed data collected from 869 patients with glomus jugulare tumors from the published literature to identify treatment variables that impacted clinical outcomes and tumor control rates. A comprehensive search of the English language literature identified 109 studies that collectively described outcomes for patients with glomus jugulare tumors. Univariate comparisons of demographic information between treatment cohorts were performed to detect differences in the sex distribution, age, and Fisch class of tumors among various treatment modalities. Meta-analyses were performed on calculated rates of recurrence and cranial neuropathy after subtotal resection (STR), gross-total resection (GTR), STR with adjuvant postoperative radiosurgery (STR+SRS), and SRS alone. The authors identified 869 patients who met their inclusion criteria. In these studies, the length of follow-up ranged from 6 to 256 months. Patients treated with STR were observed for 72 ± 7.9 months and had a tumor control rate of 69% (95% CI, 57% to 82%). Those who underwent GTR had a follow-up of 88 ± 5.0 months and a tumor control rate of 86% (95% CI, 81% to 91%). Those treated with STR+SRS were observed for 96 ± 4.4 months and had a tumor control rate of 71% (95% CI, 53% to 83%). Patients undergoing SRS alone had a follow-up of 71 ± 4.9 months and a tumor control rate of 95% (95% CI, 92% to 99%). The authors' analysis found that patients undergoing SRS had the lowest rates of recurrence of these 4 cohorts, and therefore, these patients experienced the most favorable rates of tumor control ($p < 0.01$). Patients who underwent GTR sustained worse rates of cranial nerve (CN) deficits with regard to CNs IX-XI than those who underwent SRS alone; however, the rates of CN XII deficits were comparable.

Section Summary

The evidence review related to the use of SRS for glomus jugulare tumors identified includes a large meta-analysis, which suggested that SRS treatment is associated with improved patient outcomes.

Hemangioblastoma

In 2015, Kano et al. published a large, multicenter international study evaluating patients with intracranial hemangioblastomas who underwent SRS.^[30] The study included 186 patients with 182 hemangioblastomas. Eighty patients had von Hippel-Lindau disease (VHL) and the rest had sporadic hemangioblastomas. The median margin dose was 18 Gy for VHL and 15 Gy for sporadic hemangioblastomas. After a median of 5 years, 29 patients had died. Survival after SRS at 3, 5, and 10 years was 94%, 90%, and 74% respectively. Local tumor control rates at 3, 5, and 10 years were 92%, 89%, and 79% respectively. Longer survival was associated with younger age, absence of neurological symptoms, fewer tumors, and higher Karnofsky Performance Status.

Sayer evaluated 14 patients with 26 hemangioblastomas (7 sporadic and 7 VHL) treated with gamma knife radiosurgery (10 received resection).^[31] The median age was 45 years and the median follow-up was 3 years. At 1, 5, and 10 years, the local tumor control rates were 89%, 74%, and 50% respectively. Patients with sporadic tumors, women, and larger tumors were more likely to experience tumor progression. In patients with multiple hemangioblastomas, the radiosurgically treated lesion was 7.9 times more likely to progress after treatment. Although the sample size is small, this study reports that patients with multiple hemangioblastomas being less likely to obtain local tumor control following radiosurgery.

Puataweepong et. al published a case series study evaluating 14 patients with 56 intracranial hemangioblastomas treated with SRS and SRT.^[32] The median dose was 20 Gy and the median follow-up was 24 months (11-89 months). The local control rates at 1, 2, and 6 years were 98%, 88%, and 73% respectively. No adverse events were reported.

Hanakita reported outcomes for 21 patients receiving SRS for 57 intracranial hemangioblastomas (7 sporadic and 14 VHL).^[33] The median margin dose was 18 Gy (range 14-20 Gy). The median follow-up was 96 months. The tumor control rates at 5 and 10 years were 67% and 44% for sporadic patients and 97% and 83% for VHL. Seven patients with 10 tumors showed progression after SRS.

A prospective study evaluated 20 VHL patients undergoing SRS for 44 CNS hemangioblastomas. Mean follow-up ranged from 3-17 years and the mean prescription dose was 18.9 Gys.^[34] All patients were alive at the last follow-up. Local control rates were 91%, 83%, 61%, and 51% at 2, 5, 10, and 15 years after SRS. Further analysis did not identify any variables with worse tumor control. No long-term adverse events were reported.

Moss et al. published a retrospective case series which evaluated 31 patients with 92 hemangioblastomas (26 patients had VHL).^[35] The mean patient age was 41 years and the average dose was 23.4 Gy. Follow-up (median 69 months) data was available for 82 tumors. The actuarial local control rates at 36 and 60 months were 85% and 82%, respectively. A total of five patients experienced radiation necrosis. The authors concluded that SRS can be alternative to surgery for patients.

Kano published a case series retrospectively evaluating 32 consecutive patients with recurrent or residual intracranial hemangioblastomas (19 sporadic and 13 VHL) who underwent SRS.^[36] Surgical resection was performed in 31 patients. At a median follow-up of 50 months, 7 patients died from disease progression. The overall survival rates after SRS were 100%, 94%, and 69% at 1, 3, and 7 years, respectively. Improved progression free survival factors were VHL disease, solid tumor, lower tumor volume, and greater marginal dose.

Many additional noncomparative older studies have been published showing similar outcomes as reported in the more recent studies above.^{[37] [38-40]}

Overall Summary

The current evidence is limited to small, nonrandomized studies. The evidence consistently demonstrates improvement in tumor control, disease-free progression, disease-free survival, and survival outcomes after the use of SRS in patients with hemangioblastomas. The largest multicenter study reported that longer survival was associated with younger age, absence of neurological symptoms, and higher Karnofsky Performance Status. In addition, multiple studies reported better outcomes in patients undergoing SRS with fewer tumors. Given that hemangioblastomas are rare tumors, there will never be

large, comparative studies comparing multiple interventions. The current evidence base consistently reported improved outcomes including overall survival and a low rate of adverse events in patients with hemangioblastomas undergoing SRS.

Hemangiopericytoma

Hemangiopericytoma (HPC) is a rare tumor type. In addition, it often recurs and has a high incidence of metastasis. The standard treatment is surgery, although SRS is the recent focus of study in the management of recurrent, metastatic, and residual hemangiopericytomas.

Veeravagu published a study evaluating SRS for the management of recurrent, metastatic, and residual HPC. The study reviewed 14 patients from the Stanford radiosurgery database who underwent radiotherapy for recurrent, metastatic, and residual HPC. All patients were previously treated with surgical resection. A total of 24 tumors were treated with a median follow-up of 37 months. Progression free survival rates for the 22 treated tumors were 95%, 71.5%, and 71.5% at 1, 3, and 5 years after radiotherapy using multiple treatments.

Kim published a case series with nine patients and 17 intracranial HPCs. The mean radiological follow-up was 34 months. All patients were treated with SRS. Tumor control was achieved in 14 lesions. Actuarial local tumor control rates at 1, 2, and 5 years were 100%, 85%, and 68%. No adverse events were found. The authors concluded that doses higher than previously used (around 15 Gy) are desirable to achieve better local tumor control.

Kano evaluated 20 patients with 296 tumors treated with previous surgical resection and undergoing SRS. The mean patient age was 52 years and 12 patients had undergone previous fractionated radiotherapy before the SRS. Sixteen patients had low-grade HPCs and 4 had high-grade anaplastic HPCs. The median marginal dose was 15 Gy. After a median follow-up of 48 months, the overall survival at 1, 3, and 5 years was 100%, 86%, and 14%. The progression free survival rate after SRS for high grade tumors was 89%, 67%, and 0% during the same timeframes. The progression free survival for low grade HPCs at 5 years was 90%. Five patients had died of metastases and 3 died of disease progression.

Liu published a retrospective case series of 26 patients with intraspinal HPC. The mean age was 34 years and all patients underwent at least one surgery and most received radiotherapy. The 5-year Kaplan-Meier rate of survival was 76% although the 5-year recurrence rate of survival was only 30%. Patients with high grade tumors had higher recurrence and shorter survival time.

A case series (Chang) evaluated eight patients with recurrent HPC who received SRS (median dose 20.5 Gy). The mean age was 45 years and all patients were previously treated with resection and 5 previous had external beam radiation. Six of the 8 tumors treated decreased in size and two progressed. There were no adverse events reported.

Ecker et al. retrospectively evaluated 38 patients with HPC in the primary central nervous system. Eighteen patients previously received surgical resection. The 5-year disease-free survival rate was 89%. No survival benefit was observed for patients that received initial adjuvant external0-beam radiation therapy. Compared to low grade tumors, high grade tumors recurred 6.7 years earlier.

Sheehan retrospectively evaluated 14 patients with recurrent HPC who underwent SRS for 15 discrete tumors. Most of the patients received previous treatments including resection (n=27) and conventional radiotherapy (n=7). The mean follow-up was 31 months and the mean radiation dose was 15 Gy. Local

recurrences occurred at a median of 21 months (12-75 months) after SRS. Survival rates and local tumor control occurred in 100% and 76% respectively. Four of 14 patients had metastases. The authors concluded that SRS did not protect against intra- or extracranial metastases.

Someya reported the outcomes for 4 patients who received surgical resection and SRS. Three patients had local recurrence at 30, 54, and 138 months and on patient at 49 months. Distant metastases were reported in 2 patients.

Overall Summary

The current evidence is limited to small nonrandomized studies. However, the current evidence base consistently demonstrates significant improvement in tumor control, disease-free progression, disease-free survival, and survival outcomes with use of SRS to treat hemangiopericytoma after surgical resection. The improved outcomes were reported on average up to 5-years after SRS although, the rate of reoccurrence increased with longer follow-up periods typically after 3 years. The overall rates of adverse events were low. Given that hemangiopericytomas are rare tumors, there will never be large comparative studies comparing multiple interventions that report on long-term outcomes.

Pituitary Adenoma

In 2013, Chen et al reported results from a systematic review and meta-analysis of studies evaluating SRS (specifically gamma-knife surgery) for the treatment of nonfunctioning pituitary adenoma that included a volumetric classification.^[41] Seventeen studies met the inclusion criteria, including 7 prospective cohort studies and 10 retrospective cohort studies, with 925 patients included in the meta-analysis. Outcomes were reported related to the rate of tumor control, rate of radiosurgery-induced optic neuropathy injury, and the rate of radiosurgery-induced endocrinologic deficits. In patients with tumor volume less than 2 mL, the rate of tumor control was 99% (95% CI, 96% to 100%), the rate of radiosurgery-induced optic neuropathy injury was 1% (95% CI, 0% to 4%), and the rate of radiosurgery-induced endocrinologic deficits was 1% (95% CI, 0% to 4%). In patients with volumes from 2 to 4 mL, the comparable rates were 96% (95% CI, 92% to 99%), 0 (95% CI, 0% to 2%), and 7% (95% CI, 2% to 14%), respectively, and in patients with volumes larger than 4 mL was 91% (95% CI, 89% to 94%), 2% (95% 0% to 5%) and 22% (95% CI, 14% to 31%), respectively. The rates of tumor control and rates of radiosurgery-induced optic neuropathy injury differed significantly across the 3 groups.

In 2014, Lee et al retrospectively reported outcomes for 41 patients treated with SRS in a cohort of 569 patients treated for nonfunctioning pituitary adenomas at 3 institutions.^[42] At a median follow-up of 48 months, on neuroimaging 34 patients (82.9%) had a decrease in tumor volume, 4 patients had tumor stability (9.8%), and 3 patients had a tumor increase (7.3%). PFS was 94% at 5 years and 85% at 10 years post-SRS. New onset or worsened pituitary deficiencies were found in 10 patients (24.4%) at a median follow-up of 52 months. The authors conclude that initial treatment with SRS for nonfunctioning pituitary adenomas may be appropriate in certain clinical settings, such as in older patients (≥ 70 years); in patients with multiple comorbidities in whom an operation would involve a high risk; in patients with clear neuroimaging and neuro-endocrine evidence of an NFA, no mass effect on the optic apparatus, and progressive tumor on neuroimaging follow-up; or in patients who wish to avoid resection.

Sheehan et al reported results from a multicenter registry of 512 patients who underwent SRS for nonfunctional pituitary adenomas.^[43] Four hundred seventy-nine (93.6%) had undergone prior resection, and 34 (6.6%) had undergone prior external beam radiotherapy (EBRT). Median follow-up was 36 months. At last follow up, 31 of 469 patients with available follow-up (6.6%) had tumor progression, leading to an actuarial PFS of 98%, 95%, 91%, and 85% at 3, 5, 8, and 10 years post-SRS, respectively.

Forty-one (9.3%) of 442 patients had worsened or new CNS deficits, more commonly in patients with tumor progression ($p=0.038$).

Section Summary

Noncomparative studies demonstrate high rates of tumor control (85% and better) for pituitary adenomas with SRS treatment, with better tumor control with smaller lesions. There is a lack of comparative studies evaluating the treatment of pituitary adenomas with SRS versus surgery or traditional radiotherapy.

Primary or Recurrent Gliomas and Astrocytomas

In a single-arm study, 11 patients with tectal gliomas were treated with Gamma Knife SRS between October 2002 and May 2011.^[44] Tectal gliomas are present in a critical location that makes surgical resection difficult, and are also commonly associated with aqueductal obstruction and consequently hydrocephalus. This necessitates some form of CSF diversion procedure before radiosurgery. Five patients had pilocytic astrocytomas and six nonpilocytic astrocytomas. Ten patients presented with hydrocephalus and underwent a CSF diversion procedure prior to SRS. The tumor volume ranged between 1.2 and 14.7 mL (median, 4.5 mL). The prescription dose was 11 to 14 Gy (median, 12 Gy). Patients were followed for a median of 40 months (range, 13-114 months). Tumor control after radiosurgery was seen in 100% of cases. In 6 of 11 cases (55%), the tumors eventually disappeared after treatment. Peritumoral edema developed in 45% of cases at an onset of 3 to 6 months after treatment. Transient tumor swelling was observed in 4 cases. Four patients developed cysts after treatment. One of these cases required aspiration and eventually disappeared, one became smaller spontaneously, and 2 remained stable.

In a retrospective review, 21 patients with recurrent malignant glioma (18 glioblastoma, 3 World Health Organization [WHO] grade III glioma), treated at initial diagnosis with surgery and standard chemoradiation, received concurrent bevacizumab (BVZ) with hypofractionated SRS (30 Gy in 5 fractions) with or without concurrent chemotherapy (temozolomide or CCNU).^[45] The median patient age was 54 years, median Karnofsky Performance Status was 80, and median target size was 4.3 mL (range, 3.4-7.5 mL). Eleven patients (52%) had previously failed BVZ. One patient had grade 3 toxicities (seizures, dysphasia), which resolved with inpatient admission and intravenous steroids/antiepileptics. Treatment-related toxicities were grade 3 ($n=1$), grade 2 ($n=9$), and grade 0-1 ($n=11$). Kaplan-Meier median PFS and OS estimates (calculated from start of SRS) for GBM patients ($n=18$) were 11.0 and 12.5 months, respectively.

In a prospective study, 15 patients with recurrent malignant glioma lesions less than 3 cm in diameter were treated with SRS in a single fraction, whereas those 3 to 5 cm in diameter received five 5-Gy fractions; BVZ was administered immediately before SRS and 2 weeks later.^[46] At initial diagnosis, patients were treated with surgery and adjuvant radiation therapy plus temozolomide and then at least 1 salvage chemotherapy regimen. The primary end point was central nervous system (CNS) toxicity. Secondary end points included survival, quality of life (QOL), microvascular properties as measured by MRI, steroid usage, and performance status. One grade 3 (severe headache) and 2 grade 2 CNS toxicities were observed. No patients experienced grade 4 to 5 toxicity or intracranial hemorrhage. Neurocognition, QOL, and Karnofsky Performance Status did not change significantly with treatment. MRI results suggested a significant decline in tumor perfusion and permeability 1 week after SRS and further decline by 2 months.

A retrospective analysis was performed on patients with recurrent malignant gliomas treated with salvage SRS from September 2002 to March 2010.^[47] All patients had experienced progression after treatment with temozolomide and radiotherapy. Salvage SRS was typically administered only after multiple post chemoradiation salvage systemic therapies had failed. Among 63 patients treated with SRS for recurrent high-grade glioma, 49 patients had WHO grade IV disease. Median follow-up was 31 months from primary diagnosis and 7 months from SRS. Median OS from primary diagnosis was 41 months for all patients. Median PFS and OS from SRS (OS-SRS) were 6 and 10 months for all patients, respectively. The 1-year OS-SRS for patients with grade IV glioma who received adjuvant (concurrent with or after SRS) bevacizumab was 50% versus 22% for patients not receiving adjuvant bevacizumab ($p=0.005$). Median PFS for patients with a WHO grade IV glioma who received adjuvant bevacizumab was 5.2 months versus 2.1 months for patients who did not receive adjuvant bevacizumab ($p=0.014$). Treatment related grade 3/4 toxicity for patients who receiving and did not receive adjuvant BVZ was 10% and 14%, respectively ($p=0.58$). On multivariate analysis, the relative risk of death and progression with adjuvant bevacizumab was 0.37 (95% CI, 0.17 to 0.82) and 0.45 (95% CI, 0.21 to 0.97). A Karnofsky Performance Status score greater than 70 and age less than 50 years were significantly associated with improved survival. The combination of salvage radiosurgery and bevacizumab to treat recurrent malignant gliomas is well tolerated and seems to be associated with improved outcomes. Prospective multiinstitutional studies are required to determine efficacy and long-term toxicity with this approach.

A total of 55 consecutive patients with high-grade glioma comprising 68 WHO grade III and WHO grade IV were treated with SRS (Gamma Knife) for local recurrences between 2001 and 2007.^[48] All patients had previously undergone microsurgery and radiochemotherapy. Complete follow-up was available in all patients, with a median follow-up of about 17 months (range, 2.5-114.2 months). Median tumor volume was 5.2 mL, prescription dose was 20 Gy (range, 14-22 Gy), and the median maximal dose was 45 Gy (range, 30-77.3 Gy). Patients with WHO grade III tumors initially showed a median survival of about 50 months, with a 2-year OS of 90%. After SRS of the recurrences, these patients showed a median survival of 24 months and a 2-year survival of 50%. The patients with WHO grade IV tumors had an initial median survival of 24 months with a 2-year survival of 51%; after the recurrence was treated with SRS, the median survival was 11 months and 2-year survival was 23%.

Section Summary

Direct evidence is not available to compare radiotherapy methods in the setting of primary or recurrent gliomas or astrocytomas. Evidence from heterogeneous observational studies demonstrates high rates of local control and survival with the use of SRS to treat gliomas in the primary and recurrent setting.

Brain Metastases

Systematic Reviews and Meta-Analyses

Roos et al examined the randomized evidence to treat brain metastases.^[49] A search of MEDLINE, EMBASE, and Cochrane databases for published papers and abstracts on relevant randomized trials was undertaken. Fourteen randomized trials were identified, 11 final reports and 3 abstracts, investigating various combinations of surgery, SRS and WBRT. Most of the trials had significant limitations. Surgery and SRS improved LC, maintenance of performance status and survival for favorable prognosis patients with solitary brain metastases relative to WBRT alone, although the absolute survival benefit for the majority was modest. Limited data suggest similar outcomes from surgery and SRS, but few patients were truly suitable for both options. For multiple (2-4) brain metastases, SRS improved LC and functional outcome but not survival. Adjuvant WBRT also improved intracranial control but not

survival; however, the neurocognitive risk: benefit ratio of WBRT was controversial. Quality-of-life data were limited.

A 2011 review by Park et al on the use of SRS for brain metastases discussed the 2 randomized trials that demonstrated that the addition of single-dose SRS to WBRT improves local tumor control and maintenance of functional status for patients.^[50] Also reviewed are 3 recent randomized trials comparing the outcomes for SRS alone versus SRS plus WBRT for limited brain metastases. All 3 trials indicated a lack of detriment in neurocognition or quality of life with the omission of WBRT, despite significantly worsened intracranial tumor control that would require additional salvage therapy in almost all patients.

A 2010 analysis, a Cochrane review,^[51] which addressed the role for both SRS and WBRT in patients with small numbers of metastatic lesions (generally no more than 3 or 4 lesions), noted that given the unclear risk of bias in the included studies, the results need to be interpreted with caution. The analysis of all included patients (3 trials) indicated that SRS plus WBRT did not show a survival benefit over WBRT alone; however, performance status and LC were significantly better in the SRS plus WBRT group

Randomized Controlled Trials

Since publication of the systematic reviews, several RCTs have been published. Chang et al concluded that patients treated with SRS plus WBRT were at a greater risk of a significant decline in learning and memory function by 4 months compared with the group that received SRS alone.^[52]

Some studies have suggested that use of radiosurgery for brain metastases should be limited to patients with 3 or fewer lesions. A randomized trial compared WBRT with WBRT plus radiosurgery boost to metastatic foci.^[53] Results stated that the significant advantage of radiosurgery boost over WBRT alone in terms of freedom from local failure did not differ among patients with 2, 3, or 4 metastases. Survival also did not depend on the number of metastases. As the number of metastases rises, so does the total volume of tissue receiving high-dose radiation, thus the morbidity risk of radiation necrosis associated with radiosurgery is likely to increase. For a large number of metastases, and for large volumes of tissue, this risk may be high enough to negate the advantage of radiosurgery plus WBRT over WBRT alone seen in patients with 4 or fewer metastases. SRS centers commonly exclude patients with more than 5 metastases from undergoing radiosurgery.^[54,55] It is difficult to identify a specific limit on the number of metastases for which the use of SRS is advantageous. A large number of very small metastases may respond to radiosurgery, as well as a small number of larger metastases.

In 2006, Aoyama et al reported on a randomized trial of SRS plus WBRT versus SRS alone for treatment of patients with 1 to 4 brain metastases.^[56] They found a 12-month intracranial tumor recurrence rate of 46.8% in the SRS plus WBRT group compared with 76.4% in the group that only received SRS. However, median survival times were not different at 7.5 and 8.0 months, respectively. They also found no differences in neurologic functional preservation. In an accompanying editorial, Raizer commented that either treatment approach is a reasonable first step, recognizing that those who select SRS alone are more likely to need subsequent salvage radiation treatments.^[57]

Nonrandomized Comparative Studies

Tian et al reported results from a retrospective, single-institution cohort study comparing neurosurgical resection to SRS for solitary brain metastases from non-small-cell lung cancer (NSCLC). Seventy-six patients were included, 38 of whom underwent neurosurgery.^[58] Median survival was 14.2 months for

the SRS group and 10.7 months for the neurosurgery group. In multivariable analysis, treatment mode was not significantly associated with differences in OS.

Noncomparative Studies

Noncomparative studies continue to evaluate the use of SRS without WBRT for the management of brain metastases and the role of SRS for the management of larger numbers of brain metastases. Yamamoto et al conducted a prospective observational study to evaluate primary SRS in patients with 1 to 10 newly diagnosed brain metastases.^[59] Inclusion criteria included largest tumor volume less than 10 mL and less than 3 cm in the longest diameter, a total cumulative volume of 15 mL or less, and a Karnofsky Performance Status score of 70 or higher. Among total 1194 patients, the median OS after SRS was 13.9 (95% CI, 12.0 to 15.6) in the 455 patients with 1 tumor, 10.8 months (95% CI, 9.4 to 12.4) in the 531 patients with 2 to 4 tumors, and 10.8 months (95% CI, 9.1 to 12.7) in the 208 patients with 5 to 10 tumors.

Rava et al, in a cohort study including 53 patients with at least 10 brain metastases, described the feasibility of SRS treatment.^[60] Median survival was 6.5 months in this cohort. Raldow et al, in a cohort of 103 patients with at least 5 brain metastases who were treated with SRS alone, demonstrated a median OS of 8.3 months, comparable with historical controls.^[61] OS was similar for patients with 5 to 9 and with at least 10 metastases (7.6 months and 8.3 months, respectively).

Yomo et al reported outcomes for 41 consecutive patients with 10 or fewer brain metastases from NSCLC who received SRS as primary treatment.^[62] The study reported 1- and 2-year OS rates of 44% and 17%, respectively, with a median survival time of 8.1 months. Distant brain metastases occurred in 44% by 1 year, with 18 patients requiring repeat SRS, 7 requiring WBRT, and 1 requiring microsurgery.

Section Summary

For cases of brain metastases, evidence from RCTs and systematic reviews indicate that the use of SRS improves outcomes in the treatment of brain metastases. SRS appears to be feasible in the treatment of larger numbers (eg, >10) of brain metastases, and outcomes after SRS treatment do not appear to be worse for patients with larger numbers of metastases, at least for patients with 10 or fewer metastases.

Uveal Melanoma

The literature on the use of SRS for uveal melanoma consists of case series; no studies directly comparing SRS with other, accepted radiation modalities used to treat uveal melanoma (brachytherapy, proton beam) are identified.

A 2012 review article summarizes the literature on the use of SRS for uveal melanoma, with long-term tumor control rates using the Gamma Knife reported to be around 90%.^[63] Initial studies using SRS for uveal melanoma reported secondary adverse effects from radiation to be common; however, more recent studies have reported lower incidences with lower total radiation doses.^[63]

The largest study to date consisted of 212 patients with choroidal melanoma, who were not suitable for brachytherapy or resection.^[64] Patients in the study received different doses of radiation, ranging from 50 Gy to 70 Gy, in 5 fractions over 7 days. Ophthalmologic examination was performed at baseline and every 3 months in the first 2 years, every 6 months until 5 years, and once a year until 10 years after SRS. The study included measurement of tumor dimension and height using standardized methods, assessment of visual acuity and routine ophthalmologic examinations. Local tumor control was 96% at 5

years, and 93% at 10 years. Thirty-two patients developed metastases, and 22 of these patients died during the follow-up period. Median visual acuity decreased from 0.55 at baseline to hand motion ($p < 0.001$). The authors concluded that SRS was sufficient to achieve excellent local tumor control in patients with melanoma of the choroid, and that disease outcome and vision were comparable with that achieved with proton beam radiotherapy.

Since publication of the 2012 review, several studies have reported outcomes from SRS for intraocular melanoma. Wackernagel et al reported outcomes for 189 patients with choroidal melanoma treated with SRS (Gamma Knife).^[65] All patients with choroidal melanoma at the authors' institution were offered SRS as an alternative to enucleation if they wished to retain their eye, and other globe-preserving treatment options were not feasible because of tumor size or location or the patient's general health. Sixty-six patients (37.3%), all treated before 2003, received high-dose SRS (35-80 Gy); subsequently, all patients received low-dose SRS (30 Gy in 87 patients and 25 Gy in 24 patients). The median overall follow-up was 39.5 months. During follow-up, local tumor control was achieved in 167 patients (94.4%). Enucleation was required in 25 patients, 7 due to tumor recurrence and 18 due to radiation-induced adverse effects. OS and distant metastasis rates are not reported.

Furdova et al reported outcomes for a cohort of 96 patients who underwent SRS at a single center in Slovakia for stage T2/T3 uveal melanoma.^[66] Local tumor control occurred in 95% of patients at 3 years of follow-up and in 85% of patients at 5 years of follow-up. Eleven patients (11.5%) required secondary enucleation between 3 and 5 years post-SRS due to radiation neuropathy or secondary glaucoma.

Additional case series using SRS for uveal melanoma have suggested that SRS is a possible eye-sparing option for patients, with outcomes comparable with enucleation or other radiation modalities.^[67-69]

Section Summary

The current evidence for uveal melanoma is limited to small nonrandomized studies. However, the current evidence base consistently demonstrates significant improvement in tumor control and survival outcomes. The improved outcomes were reported on average up to 3-5 years after SRS.

STEREOTACTIC BODY RADIATION THERAPY (Stereotactic Ablative Radiotherapy)

Spinal Tumors

Gerszten et al reported on the outcomes of 115 patients with spinal tumors of varying etiologies, ie, benign, metastatic, single, or multiple lesions, in a variety of locations, ie, cervical, thoracic, lumbar, sacral, who were treated with the CyberKnife in a single session.^[70] Most patients were treated for pain control and also had prior EBRT. The authors point out that radiotherapy of the spinal cord is limited by its low tolerance and that if a radiation dose could be targeted more accurately at the lesions, higher doses could be delivered in a single fraction. They further point out that conventional methods of delivering intensity-modulated radiation therapy (IMRT) are limited due to lack of target immobilization. Axial and radicular pain improved in 74 of the 79 symptomatic patients. There was no acute radiation toxicity or new neurologic deficits. Conventional EBRT typically is delivered over a course of 10 to 20 fractions. In contrast, in this study, only 1 CyberKnife treatment session was used. In a 2005 study, Degen et al reported on the outcomes of 51 patients with 72 spinal lesions who were treated with the CyberKnife.^[71] Patients underwent a median of 3 treatments. Pain was improved, as

measured by declining mean visual analog scale (VAS) score, and quality of life was maintained during the 1-year study period.

Gerszten et al recently published results on a series of 500 cases from a single institution (334 tumors had previously undergone external beam irradiation) using the CyberKnife system.^[72] In this series, the maximum intratumoral dose ranged from 12.5 Gy to 25 Gy, with a mean of 20 Gy. Long-term pain improvement occurred in 290 of 336 cases (86%). Long-term radiographic tumor control was demonstrated in 90% of lesions treated with radiosurgery as a primary treatment modality. Twenty-seven of 32 cases (84%) with a progressive neurologic deficit before treatment experienced at least some clinical improvement. Chang et al reported on phase 1/2 results of SBRT in 74 spinal lesions in 63 patients (55% had prior irradiation) with cancer.^[73] The actuarial 1-year tumor progression-free incidence was 84%. Pattern-of-failure analysis showed 2 primary mechanisms of failure: recurrence in the bone adjacent to the site of previous treatment and recurrence in the epidural space adjacent to the spinal cord. The authors concluded that analysis of the data obtained in their study supports the safety and effectiveness of SBRT in cases of metastatic spinal tumors. They add that they consider it prudent to routinely treat the pedicles and posterior elements using a wide bone margin posterior to the diseased vertebrae because of the possible direct extension into these structures and for patients without a history of radiotherapy, more liberal spinal cord dose constraints than those used in the study.

Sahgal et al evaluated rates of vertebral compression fractures after SBRT in 252 patients with 410 spinal segments treated with SBRT.^[74] Fifty-seven fractures were observed (13.9% of spinal segments treated), with 27 *de novo* fractures and 30 cases of existing fracture progression. Most fractures occurred relatively early posttreatment, with a median and mean time to fracture of 2.46 months and 6.33 months, respectively. Radiation dose per fraction, baseline vertebral compression fracture, lytic tumor, and baseline spinal misalignment were predictive of fracture risk.

Section Summary

SBRT has been shown to improve outcomes (reduce pain) in patients with spinal (vertebral) tumors. Most of the literature addresses metastases that recur after prior radiotherapy.

Non-Small-Cell Lung Cancer

Systematic Reviews

In 2014, Zheng et al reported results from a systematic review and meta-analysis comparing survival after SBRT with survival after surgical resection for the treatment of stage I NSCLC.^[75] The authors included 40 studies reporting outcomes from SBRT, including 4850 patients, and 23 studies reporting outcomes after surgery published in the same time period, including 7071 patients. For patients treated with SBRT, the mean unadjusted OS rates at 1, 3, and 5 years were 83.4%, 56.6%, and 41.2%, respectively. The mean unadjusted OS rates at 1, 3, and 5 years were 92.5%, 77.9%, and 66.1%, respectively, with lobectomy, and 93.2%, 80.7%, and 71.7% with limited lung resections. After adjustment for surgical eligibility (for the 27 SBRT studies that reported surgical eligibility) and age, in a multivariable regression model, the treatment modality (SBRT vs surgical therapy) was not significantly associated with OS ($p=0.36$).

A review by Nguyen et al^[76] cites a number of studies of SBRT for early-stage lung cancer receiving a biologic equivalent dose of 100 Gy or more. Three of the studies cited reported 5-year survival that ranged from 30% to 83%; in the largest series of 257 patients, the 5-year survival was 42%. Koto et al reported on a phase 2 study of 31 patients with stage 1 NSCLC.^[77] Patients received 45 Gy in 3

fractions, but those with tumors close to an organ at risk received 60 Gy in 8 fractions. With a median follow-up of 32 months, the 3-year OS was 72%, while disease-free survival was 84%. Five patients developed grade 2 or greater pulmonary toxicity. While comparative studies were not identified, older studies have reported 3-year disease-specific survival rates of 49% for those with stage 1 disease.^[78]

Nonrandomized Comparative Studies

In a matched-cohort study design, Crabtree et al retrospectively compared outcomes between SBRT and surgical therapy in patients with stage 1 NSCLC.^[79] Four hundred fifty-eight patients underwent primary surgical resection, and 151 were treated with SBRT. Surgical and SBRT patients differed significantly on several baseline clinical and demographic characteristics, with SBRT patients having an older mean age, higher comorbidity scores, a greater proportion of peripheral tumors, and worse lung function at baseline. For the surgical group, 3-year OS and disease-free survival (DFS) were 78% and 72%, respectively. Of note, among the 458 patients with clinical stage I lung cancer, 14.8% (68/458) were upstaged at surgery and found to have occult N1 or N2 disease. For patients with occult nodal disease, 3-year and 5-year OS were 66% and 43%, respectively. For patients without occult nodal disease, 3- and 5-year OS were 80% and 68%, respectively. For the SBRT group, 3-year OS and DFS were 47% and 42%, respectively.

In a propensity score-matched analysis, 56 patients were matched based on clinical characteristics, including age, tumor size, ACE comorbidity score, forced expiratory volume in 1 second (FEV₁) percent, and tumor location (central vs peripheral). In the final matched comparison, 3-year OS was 52% versus 68% for SBRT and surgery, respectively (p=0.05), while DFS was 47% versus 65% (p=0.01). Two-, 3-, 4-, and 5-year local recurrence-free survival for SBRT was 91%, 91%, 81%, and 40%, respectively, versus 98%, 92%, 92%, and 92% for surgery (p=0.07).

Jepperson et al compared SBRT with conventional radiotherapy for patients with medically inoperable NSCLC (T1-2N0M0).^[80] The study included 100 subjects treated with SBRT and 32 treated with conventional radiotherapy. At baseline, the SBRT-treated patients had smaller tumor volume, lower FEV₁, and a greater proportion of T1 stage disease. Median OS was 36.1 months versus 24.4 months for SBRT and conventional radiotherapy, respectively (p=0.015). Local failure-free survival rates at 1 year were 93% in the SBRT group versus 89% in the conventional radiotherapy group and at 5 years 69% versus 66%, SBRT and conventional radiotherapy, respectively (p=0.99).

Port et al compared SBRT with wedge resection for patients with clinical stage IA NSCLC using data from a prospectively maintained database.^[81] One hundred sixty-four patients were identified, 99 of whom were matched based on age, sex, and tumor histology. Thirty-eight patients underwent a wedge resection only, 38 patients underwent a wedge resection with brachytherapy, and 23 patients had SBRT. SBRT patients were more likely to have local or distant recurrences than surgically-treated patients (9% vs 30%, p=0.016), but there were no differences between the groups in disease-free 3-year survival (77% for wedge resection vs 59% for SBRT, p=0.066).

Varlotta et al compared surgical therapy (n=132 with lobectomy, n=48 with sublobar resection) with SBRT (N=137) in the treatment of stage I NSCLC.^[82] Mortality was 54% in the SBRT group, 27.1% in the sublobar resection group, and 20.4% in the lobar resection group. After matching for pathology, age, sex, tumor diameter, aspirin use, and Charlson Comorbidity Index, patients with SBRT had lower OS than patients treated with either wedge resection (p=0.003) or lobectomy (p<0.000).

Noncomparative Studies

Timmerman et al evaluated the toxicity and efficacy of SBRT in a high-risk population of patients with early stage but medically inoperable lung cancer.^[83] in a phase 2 North American multicenter study of patients aged 18 years or older with biopsy-proven peripheral T1-T2N0M0 non-small-cell tumors (<5 cm in diameter) and medical conditions precluding surgical treatment. The prescription dose was 18 Gy per fraction × 3 fractions (54 Gy total), with the entire treatment lasting between 1.5 to 2 weeks. The primary end point was 2-year actuarial primary tumor control; secondary end points were DFS (ie, primary tumor, involved lobe, regional, and disseminated recurrence), treatment-related toxicity, and OS. A total of 59 patients accrued, 55 of whom were evaluable (44 patients with T1 tumors, 11 patients with T2 tumors) with a median follow-up of 34.4 months (range, 4.8-49.9 months). Only 1 patient had primary tumor failure; the estimated 3-year primary tumor control rate was 97.6% (95% CI, 84.3% to 99.7%). Three patients had recurrence within the involved lobe; the 3-year primary tumor and involved lobe (local) control rate was 90.6% (95% CI, 76.0% to 96.5%). Two patients experienced regional failure; the local-regional control rate was 87.2% (95% CI, 71.0% to 94.7%). Eleven patients experienced disseminated recurrence; the 3-year rate of disseminated failure was 22.1% (95% CI, 12.3% to 37.8%). The rates for DFS and OS at 3 years were 48.3% (95% CI, 34.4% to 60.8%) and 55.8% (95% CI, 41.6% to 67.9%), respectively. The median OS was 48.1 months (95% CI, 29.6 months to not reached). Protocol-specified treatment-related grade 3 adverse events were reported in 7 patients (12.7%; 95% CI, 9.6% to 15.8%); grade 4 adverse events were reported in 2 patients (3.6%; 95% CI, 2.7% to 4.5%). No grade 5 adverse events were reported. The authors concluded that patients with inoperable NSCLC who received SBRT had a survival rate of 55.8% at 3 years, high rates of local tumor control, and moderate treatment-related morbidity.

Hof et al reported on outcomes (median follow-up, 15 months) for 42 patients with stages I and II lung cancer who were not suitable for surgery and who were treated with stereotactic radiotherapy.^[84] In this series, at 12 months, OS was 75% and DFS was 70%. Better local control was noted with higher doses of radiation.

In a prospective evaluation of 185 medically inoperable patients with early (T1-T2N0M0) NSCLC treated with SBRT, Allibhai et al evaluated the influence of tumor size on outcomes.^[85] Over a median follow-up of 15.2 months, tumor size (maximum gross tumor diameter) was not associated with local failure but was associated with regional failure ($p=0.011$) and distant failure ($p=0.021$). Poorer OS ($p=0.001$), DFS ($p=0.001$), and cause-specific survival ($p=0.005$) were also significantly associated with tumor volume more significant than diameter.

Harkenrider et al reported outcomes after SBRT for 34 patients with unbiopsied lung cancer, with estimated rates of 2-year regional control, distant control, and OS of 80%, 85%, and 85%, respectively.^[86]

Section Summary

Although no comparative data are available, studies have shown that SBRT for patients with stage 1 NSCLC who are not candidates for surgical resection because of comorbid conditions or for those with early stage disease who refuse surgery, survival rates may be comparable with surgical resection. Therefore, SBRT may be considered medically necessary in patients with stage T1 and T2a NSCLC (not larger than 5 cm in diameter) showing no nodal or distant disease.

Hepatocellular Carcinoma

Systematic Reviews and Meta-Analyses

Meng et al. (2009) conducted a systematic review and meta-analysis of transcatheter arterial chemoembolization (TACE) in combination with radiotherapy compared with TACE alone for unresectable hepatocellular carcinoma (HCC) using meta-analysis of data from the literature involving available trials.^[87] Seventeen trials involving 1476 patients were identified. Five were RCTs, and 12 were non-RCTs. In terms of quality, 5 RCTs were graded B, and the 12 nonrandomized studies were graded C. Results showed that TACE plus radiotherapy significantly improved survival and tumor response over TACE alone. The authors concluded that considering the strength of the evidence, additional RCTs are needed before combination TACE and radiotherapy can be routinely recommended.

A 2012 systematic review conducted by Tao and Yang, assessed the efficacy and safety of SBRT for treating primary and secondary hepatic neoplasms.^[88] The review included prospective clinical trials published in English. Fifteen studies involving 158 patients with primary tumors and 341 patients with metastases to the liver were included. Treatment was performed in 1 to 10 fractions to total doses of 18 to 60 Gy. Most studies that were included reported outcomes for patients with both primary and metastatic disease, without separating out outcome data for primary tumors only. In addition, some studies reported on outcomes for primary liver tumors including cholangiocarcinomas. At Indiana University, in a phase I study, Cardenes et al treated 17 HCC patients with Child-Turcotte-Pugh (CTP) CTP-A or CTP-B, 1 to 3 lesions and cumulative tumor diameter of 6 cm or less. Patients with CTP-A were treated in 3 fractions with the dose escalated from 12 to 16 Gy. For patients with CTP-B, the dose was modified to 5 fractions starting at 8 Gy per fraction and was not escalated because 2 patients treated at 3 × 14 Gy developed grade 3 hepatic toxicity. The 1-year OS was 75%, and there were no local failures during the median 24 months of follow-up.

Building on the Phase 1 study, 36 patients with CTP-A disease were treated with 3×18 Gy, and 24 patients with CTP-B disease were treated with 5×8 Gy. With this regimen, Andolino et al reported complete response, partial response, and stable disease for 30%, 40%, and 25% of tumors, respectively. Two-year local control, PFS, and OS were 90%, 48%, and 67%, respectively, with a median PFS of 20.4 months and OS of 44.4 months.

In an attempt to extend the use of SBRT to larger lesions, Shin et al treated 6 patients with large tumors (median tumor volume, 1288 mL; range, 1008-1815 mL) with no worse than CTP-A liver disease and without extrahepatic metastases. The 4 × 8–10 Gy regimen was relatively safe with only 1 case of grade 3 changes in transaminases. However, 1-year OS was only 33%, in part due to advanced disease. One-year LC and OS rates were 50% to 100% and 33% to 100%, respectively. There were 13 cases of radiation-induced liver disease and 4, grade 5; 6, grade 4; and 69, grade 3 adverse events reported.

Noncomparative Studies

Scorsetti published a single center case series study with 43 patients (63 HCC lesions). Median follow-up was 8 months (range 3-43 months).^[89] Median OS was 18 months +/-5.8 months. Actuarial local control was 64.4% +/-11.5% at 24 months. Overall survival was correlated with local control and gross tumor volume less than 5 cm. No radiation induced liver disease was reported. Several patients experienced significant toxicity (≥ Grade 3).

Bujold et al reported on sequential phase 1 and 2 trials of SBRT for locally advanced HCC.^[90] Two trials of SBRT for patients with HCC who were considered to be unsuitable for standard locoregional therapies were conducted from 2004 to 2010. All of the patients had CTP class A disease. The primary end points were toxicity and LC at 1 year, defined as no progressive disease of irradiated HCC by

RECIST (Response Evaluation Criteria in Solid Tumors). A total of 102 patients were evaluable (n=50 in trial 1 from 2004-2007; n=52 in trial 2 from 2007-2010). Underlying liver disease was hepatitis B in 38% of patients, hepatitis C in 38%, alcohol-related in 25%, and other in 14%, and none in 7%. Fifty-two percent received prior therapies (excluding sorafenib). TNM stage was III in 66% of patients, and 61% had multiple lesions. Median gross tumor volume was 117.0 mL (range, 1.3-1913.4 mL). Tumor vascular thrombosis (TVT) was present in 55% and 12% of patients had extrahepatic disease. LC at 1 year was 87% (95% CI, 78% to 93%). Toxicity of grade 3 or more was seen in 30% of patients. In 7 patients (2 with TVT and progressive disease), death was possibly related to treatment (1.1-7.7 months after SBRT). Median OS was 17.0 months (95% CI, 10.4 to 21.3 months).

Andolino et al evaluated the safety and efficacy of SBRT for the treatment of primary HCC.^[91] From 2005 to 2009, 60 patients with liver-confined HCC were treated with SBRT: 36 CTP class A and 24 CTP class B. The median number of fractions, dose per fraction, and total dose was 3, 14 Gy, and 44 Gy, respectively, for those with CTP class A cirrhosis and 5, 8 Gy and 40 Gy, respectively, for those with CTP class B. The records of all patients were reviewed, and treatment response was scored according to RECIST v1.1. Toxicity was graded according to the Common Terminology Criteria for Adverse Events v4.0. LC, time to progression (TTP), PFS, and OS were calculated according to Kaplan-Meier method. The median follow-up time was 27 months, and the median tumor diameter was 3.2 cm. The 2-year LC, PFS, and OS were 90%, 48%, and 67%, respectively, with median TTP of 47.8 months. Subsequently, 23 patients underwent transplant, with a median time to transplant of 7 months. There were no grade 3 or greater nonhematologic toxicities. Thirteen percent of patients experienced an increase in hematologic/hepatic dysfunction greater than 1 grade, and 20% experienced progression in CTP class within 3 months of treatment. The authors concluded that SBRT is a safe, effective, noninvasive option for patients with HCC of 6 cm or less and that SBRT should be considered when bridging to transplant or as definitive therapy for those ineligible for transplant.

Ibarra et al evaluated tumor response to SBRT in a combined multicenter database.^[92] Patients with advanced HCC (n=21) or intrahepatic cholangiocarcinoma (ICC, n=11) treated with SBRT from 4 academic medical centers were entered into a common database. Statistical analyses were performed for freedom from local progression (FFLP) and patient survival. The overall FFLP for advanced HCC was 63% at a median follow-up of 12.9 months. Median tumor volume decreased from 334.2 to 135 cm³ (p<0.004). The median time to local progression was 6.3 months. The 1- and 2-year OS rates were 87% and 55%, respectively. The incidence of grade 1 to 2 toxicities, mostly nausea and fatigue, was 39.5%. Grade 3 and 4 toxicities were present in 2 and 1 patients, respectively.

Price et al reported the results of a Phase 1/2 trial that evaluated the radiologic response in 26 patients with HCC who were not surgical candidates and were treated with SBRT between 2005 and 2008.^[93] Eligibility criteria included solitary tumors of 6 cm or less or up to 3 lesions with sum diameters of 6 cm or less, and well-compensated cirrhosis. All patients had imaging before, at 1 to 3 months, and every 3 to 6 months after SBRT. Patients received 3 to 5 fractions of SBRT. Median SBRT dose was 42 Gy (range, 24-48 Gy). Median follow-up was 13 months. Per RECIST, 4 patients had a complete response (CR), 15 had a partial response (PR), and 7 achieved stable disease (SD) at 12 months. One patient with SD experienced progression marginal to the treated area. The overall best response rate (CR + PR) was 73%. In comparison, by European Association for the Study of the Liver (EASL) criteria, 18 of 26 patients had 50% or more nonenhancement at 12 months. Thirteen of 18 demonstrated 100% nonenhancement, being greater than 50% in 5 patients. Kaplan-Meier 1- and 2-year survival estimates were 77% and 60%, respectively. SBRT is effective therapy for patients with HCC with an overall best response rate (CR + PR) of 73%.

Louis et al evaluated the feasibility, tolerance, and toxicity of SBRT in 25 HCC patients who were not eligible for other treatment modalities.^[94] All patients had liver cirrhosis with an Eastern Cooperative Oncology Group performance score of less than 2 and pretreatment Child scores ranging from A5 to B9. A total dose of 45 Gy in 3 fractions of 15 Gy each was prescribed to the 80% isodose line (95% of the planning target volume [PTV] received 45 Gy) and delivered to the target volume over 10 to 12 days. Overall, the treatment was well tolerated with 2 grade 3 acute toxicities and no acute grade 4 toxicities. Late toxicity was minimal; all observed late toxicities occurred within the first 6 months of follow-up. Three hepatic recurrences at a distance from the initial target were observed. The actuarial 1- and 2-year LC rate was 95% (95% CI, 69% to 95%). At a median overall follow-up of 12.7 months (range, 1-24 months), 6 of the 25 (24%) patients have died. Overall actuarial survival at 1 and 2 years was 79% (95% CI, 52% to 92%) and 52% (95% CI, 19% to 78%), respectively.

Kwon et al evaluated the long-term effect of SBRT for primary HCC in 42 patients ineligible for local ablation therapy or surgical resection.^[95] Median tumor volume was 15.4 cc (3.0-81.8), and the median follow-up duration was 28.7 months (8.4-49.1). CR for the in-field lesion was initially achieved in 59.6% and partial response (PR) in 26.2% of patients. Hepatic out-of-field progression occurred in 18 patients (42.9%) and distant metastasis developed in 12 (28.6%) patients. Overall 1-year and 3-year survival rates were 92.9% and 58.6%, respectively. In-field PFS at 1 and 3 years was 72.0% and 67.5%, respectively. Patients with smaller tumors had better in-field PFS and OS rates (<32 cc vs ≥32 cc, p<0.05). No major toxicity was encountered, but 1 patient died with extrahepatic metastasis and radiation-induced hepatic failure.

Yoon et al reported outcomes for 93 patients with primary nonmetastatic HCC treated with SBRT at a single institution.^[96] The median follow-up was 25.6 months. OS at 1 and 3 years was 86% and 53.8%, respectively. The main cause of treatment failure was intrahepatic (ie, out-of-field) metastases. At 1 and 3 years, LC rates were 94.8% and 92.1%, respectively, and distant metastasis-free survival rates were 87.9% and 72.2%, respectively. However, intrahepatic recurrence-free survival rates at 1 and 3 years were 51.9% and 32.4%, respectively.

Jung et al reported rates of radiation-induced liver disease in patients with HCC treated with SBRT for small (<6 cm), nonmetastatic HCC that was not amenable to surgery or percutaneous ablative therapy.^[97] Ninety-two patients were included, 17 of whom (18.5%) developed grade 2 or worse radiation-induced liver disease within 3 months of SBRT. In multivariable analysis, Child-Pugh class was the only significant predictor of radiation-induced liver injury. The 1- and 3-year survival rates were 86.9% and 54.4% respectively; with the median survival of 53.6 months. The presence of radiation-induced liver disease was not associated with survival.

Section Summary

The current evidence base is largely heterogenous and includes mostly prospective cohort studies that report outcomes for patients with HCC. Many of the studies were conducted on patients eligible for transplant or who were not eligible for other treatment modalities. Local control and overall survival among the study participants were generally over 70% at 1-3-years follow-up. Studies reported a reduction in these outcomes after 2-3 years follow-up. Multiple studies reported better outcomes when tumors were 6 cm or less. It is important to note that multiple studies reported severe adverse events (≥ grade 3) after SBRT for a small number of study participants.

Prostate Cancer

Nonrandomized Comparative Studies

Katz et al compared quality of life (QOL) after either radical prostatectomy (n=123) or SBRT (n=216) in patients with early stage prostate cancer.^[98] QOL was assessed using the Expanded Prostate Cancer Index Composite (EPIC), addressing urinary, sexual and bowel function. The EPIC data from the SBRT group was compared with the surgery group at baseline, 3 weeks, 5, 11, 24 and 36 months (SBRT group) and baseline, 1, 6, 12, 24, and 36 months (surgery group). The largest differences in QOL occurred 1 to 6 months after treatment, with larger declines in urinary and sexual QOL occurring in the surgery group, but a larger decline in bowel QOL after SBRT. The long-term urinary and sexual QOL declines remained clinically significantly lower for the patients who underwent prostatectomy but not for the SBRT patients.

In 2014, Yu et al compared toxicities after treatment with either SBRT (N=1335) or IMRT (N=2670) as primary treatment for prostate cancer, using claims data for Medicare beneficiaries.^[99] The authors identified early stage prostate cancer patients aged 66 to 94 years treated from January 2008 to June 2011 who received either IMRT (N=53,841) or SBRT (N=1335) as primary treatment. SBRT patients were matched in a 2:1 manner based on potential confounders. SBRT was associated with higher rates of genitourinary (GU) toxicity. By 6 months after treatment initiation, 15.6% of SBRT patients had a claim indicative of treatment-related GU toxicity versus 12.6% of IMRT patients (odd ratio [OR]=1.29; 95% CI, 1.05 to 1.53; p=0.009). By 12 months posttreatment, 27.1% of SBRT versus 23.2% of IMRT patients had a claim indicative of GU toxicity (OR=1.23; 95% CI, 1.03 to 1.43; p=0.01), and by 24 months after treatment initiation, 43.9% of SBRT versus 36.3% of IMRT patients had a claim indicative of GU toxicity (OR=1.38; 95% CI, 1.12 to 1.63; p=0.001). At 6 months posttreatment, there was increased gastrointestinal (GI) toxicity for patients treated with SBRT, with 5.8% of SBRT patients having had a claim indicative of GI toxicity versus 4.1% of IMRT patients (OR=1.42; 95% CI, 1.00 to 1.85; p=0.02), but at 12 and 24 months posttreatment, there were no significant differences in GI toxicity between groups.

Noncomparative Studies

Multiple cohort studies report outcomes for patients treated with a standard dose of SRS, or for groups of patients treated with SRS at escalating doses.

McBride et al reported on a multi-institutional experience with SBRT for early stage, low-risk prostate adenocarcinoma.^[100] A total of 4 centers and 45 patients were enrolled in a phase 1, multi-institutional trial. Thirty-four patients received 7.5 Gy delivered in 5 fractions, 9 patients received 7.25 Gy delivered in 5 fractions, and 2 patients received other regimens. The variables evaluated were biochemical PFS (bPFS), PSA bounce, and toxicities. Health-related quality of life was evaluated using the Sexual Health Inventory for Men (SHIM), American Urological Association (AUA), and Expanded Prostate Cancer Index Composite (EPIC) questionnaires. The median follow-up for surviving patients was 44.5 months (range, 0-62 months). The bPFS rate at 3 years was 97.7%. The median PSA declined from 4.9 ng/mL at diagnosis to 0.2 ng/mL at last follow-up, and the median percentage PSA decline at 12 months was 80%. Nine patients experienced at least 1 PSA bounce of 0.4 ng/mL or more, and 4 patients experienced 2 PSA bounces. The median time to first PSA bounce was 11.6 months (range, 7.2-18.2 months), and the mean percentage PSA bounce was 1.07 ng/mL. There was 1 episode of late grade 3 urinary obstruction, and there were 2 episodes of late-grade 3 proctitis. There was a significant late decline in SHIM and EPIC sexual scores and a small, late decline in the EPIC Bowel domain score.

Boike et al evaluated the tolerability of escalating doses of SBRT in the treatment of localized prostate cancer.^[101] Eligible patients included those with Gleason score 2 to 6 with PSA 20 or less, Gleason score

7 with PSA 15 or less, T2b or less, prostate size 60 cm³ or less, and AUA score 15 or less. Dose-limiting toxicity was defined as grade 3 or worse GI/genitourinary (GU) toxicity by Common Terminology Criteria of Adverse Events (version 3). Patients completed QOL questionnaires at defined intervals. Groups of 15 patients received 45 Gy, 47.5 Gy, and 50 Gy in 5 fractions (45 total patients). The median follow-up is 30 months (range, 3-36 months), 18 months (range, 0-30 months), and 12 months (range, 3-18 months) for the 45 Gy, 47.5 Gy, and 50 Gy groups, respectively. For all patients, GI grade of 2 or more and grade 3 or more toxicity occurred in 18% and 2%, respectively, and GU grade 2 or more and grade 3 or more toxicity occurred in 31% and 4%, respectively. Mean AUA scores increased significantly from baseline in the 47.5-Gy dose level ($p=0.002$), as compared with the other dose levels, where mean values returned to baseline. Rectal QOL scores (Expanded Prostate Cancer Index Composite) fell from baseline up to 12 months but trended back at 18 months. In all patients, PSA control was 100% by the nadir +2 ng/mL failure definition.

Freeman and King presented the outcomes for low-risk prostate cancer patients with a median follow-up of 5 years after SBRT.^[102] Between 2003 and 2005, a pooled cohort of 41 consecutive patients from 2 institutions received SBRT for clinically localized, low-risk prostate cancer. Prescribed dose was 35 to 36.25 Gy in 5 fractions. No patient received hormone therapy. Kaplan-Meier bPFS (defined using the Phoenix method) and Radiation Therapy Oncology Group (RTOG)-toxicity outcomes were assessed. At a median follow-up of 5 years, the bPFS was 93% (95% CI, 84.7% to 100%). Acute adverse effects resolved within 1 to 3 months of treatment completion. There were no grade 4 toxicities. No late grade 3 rectal toxicity occurred, and only 1 late grade 3 GU toxicity occurred following repeated urologic instrumentation.

Jabbari et al reported PSA nadir and acute and late toxicities with SBRT as monotherapy and post-EBRT boost for prostate cancer using high-dose rate (HDR) brachytherapy fractionation.^[103] Thirty-eight patients had been treated with SBRT with a minimum follow-up of 12 months. Twenty of 38 patients were treated with SBRT monotherapy (9.5 Gy $\times$ 4 fractions), and 18 were treated with SBRT boost (9.5 Gy $\times$ 2 fractions) post-EBRT and androgen deprivation therapy. PSA nadir to date for 44 HDR brachytherapy boost patients with disease characteristics similar to the SBRT boost cohort was also analyzed as a descriptive comparison. SBRT was well-tolerated. With a median follow-up of 18.3 months (range, 12.6-43.5), 42% and 11% of patients had acute grade 2 GU and GI toxicity, respectively, with no grade 3 or higher acute toxicity to date. Two patients experienced late grade 3 GU toxicity. All patients are without evidence of biochemical or clinical progression to date, and favorably low PSA nadirs have been observed with a current median PSA nadir of 0.35 ng/mL (range, <0.01-2.1) for all patients (0.47 ng/mL; range, 0.2-2.1, for the monotherapy cohort; 0.10 ng/mL; range, 0.01-0.5, for the boost cohort). With a median follow-up of 48.6 months (range, 16.4-87.8), the comparable HDR brachytherapy boost cohort has achieved a median PSA nadir of 0.09 ng/mL (range, 0.0-3.3). The authors concluded that early results with SBRT monotherapy and post-EBRT boost for prostate cancer demonstrated acceptable PSA response and minimal toxicity; PSA nadir with SBRT boost appeared comparable with those achieved with HDR brachytherapy boost.

King et al reported the long-term outcomes of a phase 2 prospective trial of SBRT for low-risk, biopsy-proven newly diagnosed prostate cancer in 67 patients enrolled between 2003 and 2009.^[104] Low risk was defined as a prebiopsy PSA of 10 ng/mL or less, a biopsy Gleason grade of 3+3 or 3+4, and a clinical stage T1c or T2a/b. Median patient age was 66 years. Treatment consisted of 36.25 Gy in 5 fractions using SBRT with CyberKnife. Patients who had received prior therapy (eg, hormonal therapy) were excluded. The end points were early and late bladder and rectal toxicities, which were patient self-reported and graded on the RTOG scale. At baseline, 92% of patients reported no urinary issues and 8% had minor issues. Baseline function for the bowel was 89% with no issues and 11% with minor issues.

Median follow-up was 2.7 years (25th-75th percentile, 1.8-4.5 years; maximum, 5.9 years). There were no grade 4 toxicities. RTOG grade 1, 2 and 3 bladder toxicities were seen in 23%, 5% and 3% of patients, respectively. The grade 3 toxicities were attributed to dysuria exacerbated by urologic instrumentation. Grade 1, 2 and 3 rectal toxicities were seen in 12.5%, 2% and 0% of patients, respectively. There were 2 PSA, biopsy-proven failures with negative metastatic workup. The 4-year PSA relapse-free survival was 94% (95% CI, 85% to 102%). The authors concluded that significant bladder and rectal toxicities from SBRT for prostate cancer were infrequent.

A separate publication from the same phase 2 trial previously outlined reported sexual function in a subset of patients.^[105] A literature review for other radiation modalities assessed by patient self-reported questionnaires served as historical comparison. Using the EPIC-validated QOL questionnaire, the sexual function of 32 consecutive patients was analyzed at median times of 4, 12, 20, and 50 months after treatment. The median follow-up was 35.5 months (range, 12-62 months). The authors concluded that the rates of erectile dysfunction after treatment of prostate cancer with SBRT were comparable with those reported for other modalities of radiotherapy.

Katz et al^[106] performed SBRT on 304 patients with clinically localized prostate cancer (211 with high-risk disease, 81 with intermediate-risk, 12 with low-risk disease): Fifty received 5 fractions of 7 Gy (total dose, 35 Gy) and 254 received 5 fractions of 7.25 Gy (total dose, 36.25 Gy). At a median 30-month (range, 26-37 months) follow-up, there were no biochemical failures for the 35-Gy dose level. Acute grade II urinary and rectal toxicities occurred in 4% of patients with no higher grade acute toxicities. At a median 17-month (range, 8-27 months) follow-up, the 36.25-Gy dose level had 2 low- and 2 high-risk patients fail biochemically (biopsy showed 2 low- and 1 high-risk patients were disease-free in the gland). Acute grade II urinary and rectal toxicities occurred in 4.7% and 3.6% of patients, respectively. The authors concluded that the low toxicity was encouraging and that additional follow-up is needed to determine long-term biochemical control and maintenance of low toxicity.

At 6-year follow-up,^[107] late urinary grade II complications were seen in 4% of patients treated with 35 Gy and 9% of patients treated with 36.25 Gy. Five late grade III urinary toxicities occurred in patients treated with 36.25 Gy. Late grade II rectal complications were seen in 2% and 5% of patients treated with 35 Gy and 36.25 Gy, respectively. Initially, bowel and urinary QOL scores decreased but returned to baseline levels. There was an overall 20% decrease in the sexual QOL score. For patients who were potent prior to SBRT, 75% remained potent. Actuarial 5-year biochemical recurrence-free survival was 97% for patients with low-risk disease, 90.7% for those with intermediate risk, and 74.1% for high-risk patients.

Bolzicco et al reported outcomes from 100 patients treated with SBRT for localized prostate cancer, 41 of whom were low risk (PSA $\leq$ 10 ng/mL or Gleason score $\leq$ 6; or tumor category T1c-T2a), 42 were intermediate risk (PSA 10-20 ng/mL or Gleason score 7 or tumor category T2c), and 17 were high risk (PSA $>$ 20 ng/mL or Gleason score $>$ 7 or 2 median risk factors).^[108] Twenty-seven patients received androgen deprivation therapy at the discretion of their treating urologist. Sixty-two patients had acute toxicity (within the first 1-2 weeks after treatment): 34% had grade 1 and 12% grade 2 urinary toxicity; 27% had grade 1 and 18% grade 2 GI toxicity. Late urinary toxicity, primarily urgency and frequency (at 6 months or later posttreatment) occurred in 8% of the patients: 4% grade 1, 3% grade 2 and 1% grade 3. The 3-year bPFS rate was 94.4% (95% CI, 85.3% to 97.9%)

Other noncomparative studies have reported on specific outcomes after SBRT for prostate cancer, including rates of patient-reported urinary incontinence^[109] and rectal tolerance^[110] and health-related QOL outcomes.^[111]

Section Summary

Data on the use of SBRT in prostate cancer consists primarily of single-arm assessments of acute and late toxicity and early PSA outcome data retrospectively compared with historical controls. Studies have shown promising initial results on the use of SBRT in prostate cancer with seemingly low toxicity rates. One comparative study of IMRT versus SBRT from 2014 suggested higher GI and GU complication rates after SBRT; while this study had a large number of patients and attempted to control for bias using matching on observed variables, it is subject to the limitation of deriving outcome measures from claims data. Longer term follow-up is needed to assess the effect on long-term toxicities, cancer control, and patient survival..

Pancreatic Cancer

Goyal et al reported outcomes with SBRT in patients with pancreatic adenocarcinoma who were found not to be candidates for surgical resection.^[112] A prospective database of the first 20 consecutive patients receiving SBRT for unresectable pancreatic adenocarcinomas and a neuroendocrine tumor was reviewed. Mean radiation dose was 25 Gy (range, 22-30 Gy) delivered over 1 to 3 fractions. Chemotherapy was given to 68% of patients in various schedules/timing. Patients had a mean gross tumor volume (GTV) of 57.2 cm³ (range, 10.1-118 cm³) before SBRT. The mean total GTV reduction at 3 and 6 months after SBRT were 21% and 38%, respectively (p<0.05). Median follow-up was 14.57 months (range, 5-23 months). The overall rate of freedom from local progression at 6 and 12 months were 88% and 65% respectively. The probability of OS at 6 and 12 months were 89% and 56%, respectively. No patient had a complication related to fiducial markers placement regardless of modality. The rate of radiation-induced adverse events was: grade 1 to 2 (11%) and grade 3 (16%). There were no grade 4/5 adverse events seen.

Rwigema et al assessed the feasibility and safety of SBRT in patients with advanced pancreatic adenocarcinoma.^[113] The outcomes of 71 patients treated with SBRT for pancreatic cancer between 2004 and 2009 were reviewed. Forty patients (56%) had locally unresectable disease, 11 patients (16%) had local recurrence following surgical resection, 8 patients (11%) had metastatic disease, and 12 patients (17%) received adjuvant SBRT for positive margins. The median dose was 24 Gy (18-25 Gy), given in a single-fraction SBRT (n=67) or fractionated SBRT (n=4). Kaplan-Meier survival analyses were used to estimate FFLP and OS rates. The median follow-up among surviving patients was 12.7 months (4-26 months). The median tumor volume was 17 mL (5.1-249 mL). The overall FFLP rates at 6 months/1 year were 71.7% to 48.5%, respectively. Among those with macroscopic disease, FFLP was achieved in 77.3% of patients with tumor size less than 15 mL (n=22), and 59.5% for tumor size of 15 mL or more (n=37) (p=0.02). FFLP was achieved in 73% following 24 to 25 Gy, and 45% with 18 to 22 Gy (p=0.004). The median OS was 10.3 months, with 6 month/1 year OS rates of 65.3% to 41%, respectively. Grade 1-2 acute and late GI toxicity were seen in 39.5% of patients. Three patients experienced acute grade 3 toxicities. SBRT is feasible, with minimal grade 3 or more toxicity. The overall FFLP rate for all patients was 64.8%, comparable with rates with EBRT.

Chang et al reported on the local control and toxicity of SBRT for patients with unresectable pancreatic adenocarcinoma.^[52] Seventy-seven patients with unresectable adenocarcinoma of the pancreas received 25 Gy in 1 fraction. Forty-five patients (58%) had locally advanced disease, 11 patients (14%) had medically inoperable disease, 15 patients (19%) had metastatic disease, and 6 patients (8%) had locally recurrent disease. Nine patients (12%) had received prior chemoradiotherapy. Sixteen patients (21%) received between 45 to 54 Gy of fractionated radiotherapy and SBRT. Various gemcitabine-based

chemotherapy regimens were received by 74 patients (96%), but 3 patients (4%) did not receive chemotherapy until they had distant failure. The median follow-up was 6 months (range, 3-31 months) and, among surviving patients, it was 12 months (range, 3-31 months). The overall rates of FFLP at 6 months and 12 months were 91% and 84%, respectively. The 6- and 12-month isolated local recurrence rates were 5% and 5%, respectively. There was no difference in the 12-month FFLP rate based on tumor location (head/uncinate, 91% vs body/tail, 86%; $p=0.52$). The PFS rates at 6 months and 12 months were 26% and 9%, respectively. The PFS rate at 6 months was superior for patients who had nonmetastatic disease versus patients who had metastatic disease (28% vs 15%; $p=0.05$). The OS rates at 6 months and 12 months from SBRT were 56% and 21%, respectively. Four patients (5%) experienced grade 2 or greater acute toxicity. Three patients (4%) experienced grade 2 late toxicity, and 7 patients (9%) experienced grade 3 or greater late toxicity. At 6 months and 12 months, the rates of grade 2 or greater late toxicity were 11% and 25%, respectively.

Section Summary

Combined chemoradiotherapy plays a significant role in the treatment of locally advanced pancreatic cancer. The role of SBRT as a radiation technique for pancreatic tumors has not been established, and it is not clear which patients would most likely benefit. Although studies have shown promising LC rates, there have been no significant changes in patient survival compared with historical data, and some studies have shown unacceptable toxicity and questionable palliative effect.

Renal Cell Carcinoma

A 2012 systematic review on the use of stereotactic radiotherapy for primary renal cell carcinoma (RCC) identified a total of 126 patients worldwide who had been treated using this modality.^[114] A systematic search performed in January 2012 identified 7 retrospective studies and 3 prospective studies that used a wide range of techniques, doses and dose fractionation schedules. Median or mean follow-up ranged from 9 months to 57.5 months. LC was reported as 93.9% (range, 84%-100%) and the rate of severe grade 3 or higher adverse events was 3.8% (range, 0%-19%). The conclusions of the systematic review were that the current literature suggests that stereotactic radiotherapy for RCC can be delivered with good rates of LC and acceptable toxicity but that there is insufficient evidence to recommend a consensus for dose fractionation or technique, and there is a need for further prospective studies.

Beitler et al reported outcomes in 9 patients with nonmetastatic RCC, 2 of whom had bilateral RCCs.^[115] Patients were treated definitively with 40 Gy in 5 fractions using SBRT. With a median follow-up of 26.7 months, 4 of the 9 patients were alive. The survivors had a minimum follow-up of 48 months. At presentation, all 4 of the survivors had tumors of 3.4 cm or less in largest dimension, had clinically negative lymph nodes, and presented no clinical evidence of penetration of Gerota fascia or renal vein extension.

Ranck et al reported outcomes for 18 patients with RCC with limited metastases who were treated with SBRT.^[116] For patients with 5 or fewer metastatic lesions, all lesions were treated; in patients with greater than 5 lesions, rapidly-growing lesions or those close to vital organs were treated. In all, 39 metastatic lesions were treated, with a median of 2 lesions per patient. The 2-year lesion-control rate was reported as 91.4% in the 12 patients who underwent treatment for all metastases, over a median follow-up of 21.3 months. However, in these patients, 2-year freedom from new metastases was 35.7%. OS was 85% at 2 years. No patients who underwent treatment at all lesion sites died.

Section Summary

The literature on the use of SBRT for RCC consists of very small case series, which generally report high rates of LC. However, little evidence about the impact on patient outcomes can be derived from these data, nor any comparison made between this treatment modality and more established treatment modalities for RCC.

Oligometastases

The 2012 and 2013 reviews on the use of SBRT for oligometastases summarize the data on local tumor control, and in a limited subset of patients, survival, for various anatomic sites.^[117-119] The review conclusions are summarized below by type of oligometastases.

A 2012 long-term follow-up of a prospective study was reported on oligometastases treated with SBRT.^[120] The authors prospectively analyzed the long-term survival, tumor control outcomes, and freedom from widespread distant metastases (FFDM) after SBRT in 121 patients with 5 or fewer clinically detectable metastases, from any primary site, metastatic to 1 to 3 organ sites, and treated with SBRT. For patients with breast cancer, the median follow-up was 4.5 years (7.1 years for 16/39 patients alive at the last follow-up visit). The 2-year OS, FFDM and LC rate was 74%, 52%, and 87%, respectively. Six-year OS, FFDM, and LC rate were 47%, 36%, and 87%, respectively. From the multivariate analyses, the variables of bone metastases ($p=0.057$) and one versus more than one metastasis ($p=0.055$) were associated with a 4-fold and 3-fold reduced hazard of death, respectively. None of the 17 bone lesions that were from breast cancer recurred after SBRT versus 10 of 68 lesions from other organs that recurred ($p=0.095$). For patients with post breast cancers, the median follow-up was 1.7 years (7.3 years for 7 of 82 patients alive at the last follow-up visit). Two-year OS, FFDM, and LC rate were 39%, 28%, and 74%, respectively, and 6-year OS, FFDM, and LC rate were 9%, 13%, and 65%, respectively. For nonbreast cancers, a greater SBRT target volume was significantly adverse for OS ($p=0.012$) and lesion LC ($p<0.001$). Patients, whose metastatic lesions demonstrated radiographic progression after systemic therapy but before SBRT, experienced significantly worse OS compared with patients with stable or regressing disease. The authors conclude that select patients with limited metastases treated with SBRT are long-term survivors.

In 2015, Scorsetti published a case series with 82 elderly patients (111 total metastases) with oligometastases (16 localized to the abdomen, 50 liver, 45 lungs).^[121] Median age was 79 years of age. The majority of patients ($n=64$) had a single lesion and 18 had 2-4 lesions. Local response was reported for 87 lesions and local progression was reported for 24 lesions. Two-year local control findings were 76.3% $\pm$ 4.4% and overall survival was 72.0% $\pm$ 5.6%. Disease-specific survival was 81.6% $\pm$ 4.9% at 2 years. Treatment related Grade toxicity was reported. Grade 2-3 toxicity was reported in 5 patients, Grade 1 toxicity in 7 patients, and no toxicity in 85.4% of patients.

Lung Oligometastases

For isolated or a few lung metastases (including <3 or <5 , according to different selection criteria), the LC probability at 1 year has been reported in the range of 70% to 100%.^[117] The overall survival varied widely after 2-years (21%-84%) among the studies. In most case series, the most common clinical presentation is a single-lung metastasis. It is difficult to accurately evaluate survival estimates and clinical outcomes using SBRT for lung metastases due to an absence of randomized trials and because most phase 1 and 2 trials included heterogeneous patient populations.^[117]

It is also difficult to compare OS data from SBRT with that of historical surgical metastasectomy series, mainly because of the different clinical characteristics of the patients, as most patients referred for SBRT

are felt to be inoperable due to medical comorbidities that affect OS outcomes.^[117] Data from the International Registry of Lung Metastases reported OS of 70% at 2 years and 36% at 5 years in patients with a single metastasis who underwent surgical metastasectomy.^[122]

A systematic review by Siva et al on the use of SBRT for pulmonary oligometastases estimated from the largest studies included in the review a 2-year weighted OS rate of 54.5%,^[123] ranging from higher rates in a study by Norisha et al of 84%^[124] to lower rates, such as 39%, reported from a multi-institutional trial.^[125]

Since publication of the Siva et al systematic review, Osti et al reported outcomes from a prospective cohort study of SBRT for lung oligometastases.^[126] Sixty-six patients with lung oligometastases were included, most (61%) with a single pulmonary nodule. For the primary end point of LC, over a median follow-up of 14 months, LC at 1 year and 2 years was 89.1% and 82.1%, respectively. OS at 1 and 2 years was 76.4% and 31.2%, respectively, while PFS at 1 and 2 years was 53.9% and 22%, respectively. Two cases of grade 3 toxicity (pneumonitis) occurred.

Liver Oligometastases

The liver is the most common site of metastatic spread of colorectal cancer (CRC). Data show that surgical resection of limited liver metastases can result in long-term survival in select patients. However, only 10% to 20% of patients with metastatic CRC to the liver are surgical candidates. In patients who are not considered to be candidates for surgery, a variety of locally ablative techniques have been developed, the most common of which are RFA and transarterial chemoembolization. Retrospective analyses of RFA for liver metastases from CRC have shown wide variability in 5-year OS rates, ranging from 14% to 55%.^[117]

Retrospective series on the use of SBRT has reported LC rates ranging from 57% to 100% (median follow-up ranged 10 months – 4.3 years).^[117] Prospective studies have reported 1-year OS rates ranging from 61% to 85% and 2-year OS rates ranging from 30% to 62%.^[117] Another systematic review concluded similar findings evaluating similar studies.^[119] In addition, the review concluded that the rate of adverse events was low with less than 5% of patients experiencing severe toxicity (grade 3 or more).

One of the larger series that was reported by Chang et al studied outcomes of SBRT for colorectal liver metastases in a pooled patient cohort from three institutions with colorectal liver metastases.^[127] Patients were included if they had 1 to 4 lesions, received 1 to 6 fractions of SBRT, and had radiologic imaging 3 months or more posttreatment. Sixty-five patients with 102 lesions treated from 2003 to 2009 were retrospectively analyzed. Forty-seven (72%) patients had 1 or more chemotherapy regimens before stereotactic body radiotherapy, and 27 (42%) patients had 2 or more regimens. The median follow-up was 1.2 years (range, 0.3-5.2 years). The median dose was 42 Gy (range, 22-60 Gy). One- and 2-year LC rates were 67% and 55%, respectively. One- and 2-year OS rates were 72% and 38%, respectively.

These studies have had relatively short follow-up times, typically less than 18 months. They are also limited by relatively small numbers of patients in the studies and differences in the systemic therapies administered, which may affect treatment outcomes.

Adrenal Gland Oligometastases

The most frequent primary tumor that metastasizes to the adrenal glands is NSCLC. Longer OS times have been reported with resection of clinically isolated adrenal metastases when compared with

nonsurgical therapy, which has included locally ablative techniques, embolization and EBRT. LC rates at 1 year ranging from 55% to 90% have been reported, and 1-year OS rates ranging from 40% to 56% and 2-year OS ranging from 14% to 33%.^[117]

Scorsetti et al described the feasibility, tolerability and clinical outcomes of SBRT in the treatment of adrenal metastases in consecutive cancer patients.^[128] Between 2004 and 2010, a total of 34 patients, accounting for 36 adrenal metastatic lesions, were treated with SBRT. All 34 patients were clinically and radiologically evaluated during and after completion of SBRT. The following outcomes were taken into account: best clinical response at any time, LC, time-to-systemic progression, time-to-local progression, OS and toxicity. The Kaplan-Meier method was used to estimate survival and factors that could potentially affect outcomes were analyzed with Cox regression analysis. No cases of grade 3 or greater toxicity were recorded. At a median follow-up of 41 months (range, 12-75 months), 22 patients were alive. Eleven percent of lesions showed CR, 46% PR, 36% SD, and 7% progressed in the treated area. Local failure was observed in 13 cases and actuarial local control rates at 1 and 2 years were 66% and 32%, respectively. Median time-to-local progression was 19 months and median survival was 22 months.

Holy et al presented initial institutional experiences with SBRT for adrenal gland metastases.^[129] Between 2002 and 2009, 18 patients with NSCLC and adrenal metastases received SBRT for the metastatic disease. Metastases were isolated in 13 patients and multiple in 5 patients. A median PFS time of 4.2 months was seen in the entire patient group, with an increased PFS of 12 months in the 13 patients with isolated metastasis. After a median follow-up of 21 months, 77% of the patients with isolated adrenal metastasis achieved LC. In these patients, median OS was 23 months.

Casamassima et al retrospectively evaluated a single-institution's outcomes after hypofractionated SBRT for adrenal metastases.^[130] Between 2002 and 2009, 48 patients were treated with SBRT for adrenal metastases. Eight patients were treated with single-fraction SBRT and 40 patients with multifraction. Median follow-up was 16.2 months (range, 3-63 months). At time of analysis, 20 patients were alive and 28 patients were dead. One- and 2-year actuarial OS rates were 39.7% and 14.5%, respectively. The median interval to local failure was 4.9 months. The actuarial 1-year disease control rate was 9%; the actuarial 1- and 2-year LC rates were both 90%.

Chawla et al investigated the dosimetry and outcomes of patients undergoing SBRT for metastases to the adrenal glands.^[131] A retrospective review of 30 patients who had undergone SBRT for adrenal metastases from various primary sites, including lung (n=20), liver (n=3), breast (n=3), melanoma (n=1), pancreas (n=1), head and neck (n=1), and unknown primary (n=1) was performed. Of the 30 patients, 14 with 5 or fewer metastatic lesions (including adrenal) underwent SBRT, with the intent of controlling all known sites of metastatic disease. Sixteen patients underwent SBRT for palliation or prophylactic palliation of bulky adrenal metastases. Twenty-four patients had more than 3 months of follow-up with serial computed tomography. Of these 24 patients, 1 achieved CR, 15 achieved PR, 4 had SD, and 4 developed progressive disease. No patients developed symptomatic progression of their adrenal metastases. LC was poor, and most patients developed widespread metastases shortly after treatment, with 1-year survival, LC, and distant control rates of 44%, 55%, and 13%, respectively. No patient developed grade 2 or greater toxicity.

Ahmed et al reported outcomes from a single-center's experience with SBRT for treatment of metastases to the adrenal glands.^[132] Thirteen patients were included, most with lung primary tumors (n=9), with the remainder having kidney (n=2), skin (n=2), bladder (n=1), colon (n=1), and liver (n=1) as primary sites. Eleven patients (84.6%) had received prior chemotherapy since being diagnosed with metastatic

disease, and 1 patient had undergone previous SBRT to bilateral psoas muscle metastases before adrenal SBRT. At the time of analysis, 8 of 13 patients were alive. The median follow-up time for living patients was 12.3 months (range, 3.1-18 months). Median survival for the 5 patients who died was 6.9 months (range, 2.1-15.2 months). Of the 12 patients who had evaluation for LC and distant control, 11 (91.6%) had some local response to therapy, but distant failure occurred in 6 patients at a median of 2.5 months posttreatment, leading to a 1-year distant control estimate of 55%. In exploratory analysis, there was no difference between lung primary tumor and other primary tumor sites in terms of OS or distant control. Acute toxicity included grade 2 nausea in 2 patients, grade 2 abdominal pain in 1 patient, grade 1 fatigue in 5 patients, and grade 1 diarrhea in 1 patient.

Bone Oligometastases

Napieralska et al reported a series 48 cases of prostate cancer bone metastases (in 32 patients) treated with SBRT primarily for pain control.^[133] The size of the treated lesions ranged from 0.7 to 5.5 cm (mean, 3 dimension), and 31 (65%) of the treated metastases were located in the spine. At 3-month follow-up, 17 patients had complete pain relief, 2 had partial pain relief, and 2 had no pain reduction. At the end of the follow-up period, complete pain relief was observed in 28 patients and partial pain relief in 16 patients.

Spinal Oligometastases

A small number of case series have been published evaluating the use of SBRT for the treatment of spinal metastases and are summarized in the included systematic reviews addressing oligometastases. The case series are heterogenous and some include studies with mixed primary sites. The studies consistently report local control rates of 77%-93% with a median follow-up ranging from 6-21 months. Data is lacking regarding longer-term follow-up and overall survival.^[118,119]

Section Summary

The evidence for the use of SBRT to treat oligometastases is generally limited to case series with heterogenous study populations. However, the evidence consistently reports a high rate of tumor control for isolated or few metastases (≤ 3 or ≤ 5) for the liver and lung. The local tumor control is good and reported at 1-year to be in the range of 70% to 100%. The overall survival varied widely after 2-years (21%-84%) among the studies. Although some adverse events were reported, the overall rates for adverse events were low.

The evidence related to the use of SBRT for oligometastases for all other locations is heterogenous and limited to very small sample sizes, short-term follow-up, and retrospective, noncomparative analyses. More evidence is needed to establish the role of SBRT for the treatment of oligometastases for many of these locations.

Clinical Practice Guidelines

National Comprehensive Cancer Network

The National Comprehensive Network (NCCN) provides guidelines for cancer treatment by site that include the use of SRS and SBRT for certain cancers.^[134]

Cancer Site	Tumor Type	Recommendation	Version
Bone	Osteosarcoma – metastatic disease	Consider use of stereotactic radiotherapy, especially for oligometastases (category 2A)	2.2016
CNS	Adult intracranial and spinal ependymoma – spine or brain reoccurrence	- Resection with limited radiotherapy if no prior radiotherapy; consider use of SRS if geometrically favorable (category 2A) - If unresectable, radiotherapy if no prior radiotherapy; consider use of SRS if geometrically favorable (category 2A) - If progression, radiotherapy; consider use of SRS if geometrically favorable (category 2A)	1.2015
CNS	Adult medulloblastoma and supratentorial PNET – recurrent disease	If progression after localized recurrence and maximum safe resection (category 2A)	1.2015
CNS	Primary spinal cord tumors	If recurrence, radiotherapy including SRS if surgery is not possible (category 2A)	
CNS	Meningiomas – asymptomatic and small (<30 mm)	If potential neurologic consequences from surgery and if accessible, followed by RT (external beam or SRS) if WHO grade 3; consider RT resected or incompletely resected if WHO grade 2 or incompletely resected WHO grade 1 (category 2A)	1.2015
CNS	Meningiomas – asymptomatic and large (≥30 mm)	If accessible, surgery followed by RT (external beam or SRS) if WHO grade 3; consider RT for resected or incompletely resected WHO grade 1-2 (category 2A)	1.2015
CNS	Meningiomas – symptomatic and small (<30 mm)	If accessible, surgery followed by RT (external beam or SRS) if WHO grade 3 or RT (external beam or SRS) (category 2A)	1.2015
CNS	Meningiomas – symptomatic and large (≥30 mm)	If accessible, surgery followed by RT (external beam or SRS) if WHO grade 3; consider RT for resected or incompletely resected WHO grade 2 or incompletely resected WHO grade 1 or RT (external beam or SRS) (category 2A).	1.2015
CNS	Metastatic disease – 1-3 lesions, primary treatment	- If resectable, surgical resection followed by WBRT (category 1) or SRS (category 2B), OR SRS and WBRT (category 1 for 1 metastasis), OR SRS alone (category 2A) - If unresectable, WBRT or SRS (category 2A)	1.2015
CNS	Metastatic disease – 1-3 lesions, recurrence	- If local recurrence and previous surgery only, surgery, single dose or fractionated stereotactic RT, or WBRT (category 2A) - If local recurrence and previous WBRT or SRS, surgery or single dose (category 2B) or fractionated SRS (category 2A) - If distant brain recurrence and 1-3 lesions, surgery, single dose or fractionated stereotactic RT, WBRT, or consider chemotherapy	1.2015
CNS	Metastatic disease – >3 lesions, primary treatment	WBRT or SRS (category 2A). SRS can be considered for patients with good performance status and low overall tumor volume.	1.2015
CNS	Metastatic spine tumors	- If no spinal cord compression, no fracture or spinal cord instability, consider surgical stabilization or vertebral augmentation followed by RT (recommend SRS if oligometastases and radioresistant) - If no spinal cord compression, no fracture or spinal instability, RT (preferred) or chemotherapy or surgery. Consider surgery or SRS if deterioration during RT, intractable pain, or tumor progression (recommend SRS if oligometastases and radioresistant) (category 2A). - If progressive disease or recurrent disease and if previously treated with RT or surgery and RT, consider surgery or SRS (recommend SRS if oligometastases and radioresistant). - If progressive disease or recurrent disease and previously treated with chemotherapy, consider RT (recommend SRS if oligometastases and radioresistant) (category 2A).	1.2015
Colon	Metastatic to liver or lung	In patients with a limited number of liver or lung metastases, radiotherapy can be considered in highly selected cases or in the setting of a clinical trial. Radiotherapy should not be used in the	2.2016

Cancer Site	Tumor Type	Recommendation	Version
		place of surgical resection. Radiotherapy should be delivered in a highly conformal manner. The techniques can include 3-D conformal radiation therapy, IMRT, or SBRT (category 3).	
Hepatobiliary	Hepatocellular carcinoma	• If potentially resectable or transplantable, operable by performance status or comorbidity, and Child-Pugh class A or B, no portal hypertension, suitable tumor location, adequate liver reserve, and suitable liver remnant, resection if feasible or external beam radiation therapy (conformal or stereotactic) (category 2B) • If unresectable and not a liver transplant candidate, consider external beam radiation therapy (conformal or stereotactic) among other options (category 2B) • If inoperable by performance status or comorbidity and local disease with minimal or no extrahepatic disease, consider external beam radiation therapy (conformal or stereotactic) among other options (category 2B) • Palliative EBRT is appropriate for symptom control and/or prevention of complications from metastatic HCC lesions, such as bone or brain	1.2016
Lung	Non-small-cell lung cancer – Stage IA	• If negative mediastinal nodes and medically inoperable, definitive RT including stereotactic ablative radiotherapy (category 2A)	4.2016
Lung	Non-small-cell lung cancer – Stage IV, metastatic disease to single site, brain or adrenal.	• Brain metastasis: Surgical resection followed by WBRT or SRS (category 2A), or SRS and WBRT (category 1 for 1 metastasis), or SRS alone (category 2A) • If brain or adrenal metastasis and T1-2, N0-1 or T3, N0 disease: consider stereotactic ablative radiotherapy of lung lesion (category 2A) or chemotherapy followed by surgical resection of lung lesion or stereotactic ablative radiotherapy of lung lesion (category 2A)	4.2016
Pancreas	Pancreatic adenocarcinoma	• SBRT is recommended only as part of a clinical trial	2.2015
Prostate	Prostate cancer	• SBRT may be considered as a cautious alternative to conventionally fractionated RT at clinics with appropriate technology, physics, and clinical expertise	2.2016
Skin	Melanoma – metastatic	• Brain metastases: SRS either as adjuvant or primary treatment or WBRT	2.2016
Soft tissue sarcoma – extremity, superficial trunk, head/neck	Sarcoma – synchronous stage IV	• If single organ and limited tumor bulk that are amenable to local therapy: consider SBRT (category 2A) • If disseminated metastases: SBRT as a palliative option (category 2A)	1.2016
Soft tissue sarcoma – extremity, superficial trunk, head/neck	Sarcoma – recurrent disease with metastases	• If single organ and limited tumor bulk that are amenable to local therapy: consider SBRT (category 2A) • If disseminated metastases: SBRT as a palliative option (category 2A) • If isolated regional disease or nodes: consider SBRT (category 2A)	2.2014
Thyroid	Papillary, follicular, or Hurthle cell carcinoma – structurally persistent/recurrent locoregional or distant metastatic disease not amenable to radioactive iodine	• CNS metastases: for solitary lesions, either neurosurgical resection or SRS is preferred (category 2A)	2.2015

NCCN Categories

- **Category 1:** Based upon high-level evidence, there is uniform NCCN consensus that the intervention is appropriate.
- **Category 2A:** Based upon lower-level evidence, there is uniform NCCN consensus that the intervention is appropriate.
- **Category 2B:** Based upon lower-level evidence, there is NCCN consensus that the intervention is appropriate.
- **Category 3:** Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

*All recommendations are category 2A unless otherwise noted.

American College of Chest Physicians

Non-Small-Cell Lung Cancer

- In patients with stage I or II NSCLC with no medical contraindications to operative intervention, surgical resection is recommended (grade 1B-strong recommendation based on moderate evidence)^[135]
- In patients with stage I NSCLC who cannot tolerate lobectomy or segmentectomy:^[135]
 - SBRT and wedge resection are recommended over no treatment (Grade 2C).
 - SBRT is favored over wedge resection in these cases unless surgical resection may provide the benefit of definitive histologic analysis and nodal information that will result in a change in the patient's management.
 - SBRT is also favored in these patients if adequate surgical margin is unlikely with wedge resection.
- For high-risk stage I NSCLC tumors <5 cm, SBRT is preferred over conventional fractionated RT for definitive treatment when normal dose constraints can be respected.^[136]
- For tumors within 2 cm of the proximal bronchial tree, a modified SBRT treatment schedule is suggested to decrease treatment-related toxicity.^[136]
- For second primary lung cancer, SRS is an emerging technology, particularly when there is limited pulmonary reserve.^[135]

Lung Cancer

- In lung cancer patients with 1-3 brain metastases, stereotactic radiosurgery (SRS) alone is the recommended initial therapy (Grade 1A).^[137]

American Society for Radiation Oncology (ASTRO)

Central Nervous System

- Brain Metastases: SRS is recommended for the following:^[138]
 - Single brain metastases (initial management):
 - If good prognosis (expected survival 3 months or more) and complete resection possible:
 - If brain metastasis $\leq$ 3-4 cm, options include SRS and WBRT (level of evidence: I), SRS alone (level of evidence: 1), and surgery with SRS/radiation boost with or without WBRT (level of evidence: 3)

- If brain metastasis >3-4 cm, treatment options include surgery with SRS/radiation boost with or without WBRT (level of evidence: 3)
- If good prognosis and not resectable:
 - If brain metastasis ≤3-4 cm, options include SRS and WBRT (level of evidence: I), SRS alone (level of evidence: 1).
- For multiple brain metastases (initial management):
 - If good prognosis (expected survival 3 months or more) and brain metastasis ≤3-4 cm, options include SRS and WBRT (level of evidence: I), SRS alone (level of evidence: 1)

Bone Metastases

- For palliative therapy for bone metastases, ASTRO makes the following recommendations related to the use of SBRT:^[139]
 - Patients with painful bone metastases should be treated with SBRT only in clinical trials and that SBRT should not be the primary treatment of vertebral bone lesions causing cord compression.
 - For re-treatment of recurrent metastatic spine pain with SBRT, the Task Force states that the specifics of SBRT re-treatment dosing and target delineation are insufficiently defined to allow SBRT re-treatment outside of the clinical trial setting.

American Society of Clinical Oncology

The ASCO guideline addresses the management of brain metastases for patients with human epidermal growth factor receptor 2 (HER2) -positive advanced breast cancer.^[140] ASCO makes the following recommendations:

- For patients with a favorable prognosis for survival and a single brain metastasis, treatment options include surgery with postoperative radiation, stereotactic radiosurgery (SRS), whole-brain radiotherapy (WBRT; SRS), fractionated stereotactic radiotherapy (FSRT), and SRS (WBRT), depending on metastasis size, resectability, and symptoms. After treatment, serial imaging every 2 to 4 months may be used to monitor for local and distant brain failure.
- For patients with a favorable prognosis for survival and limited (two to four) metastases, treatment options include resection for large symptomatic lesion(s) plus postoperative radiotherapy, SRS for additional smaller lesions, WBRT (SRS), SRS (WBRT), and FSRT for metastases 3 to 4 cm.
- For metastases 3 to 4 cm, treatment options include resection with postoperative radiotherapy. In both cases, available options depend on resectability and symptoms.
- For patients with progressive intracranial metastases despite initial radiation therapy, options include SRS, surgery, WBRT, a trial of systemic therapy, or enrollment onto a clinical trial, depending on initial treatment. For patients in this group who also have diffuse recurrence, best supportive care is an additional option.

OVERALL SUMMARY

Stereotactic Radiosurgery

SRS is an established safe and effective treatment modality for many benign and malignant intracranial tumors/conditions. The evidence, largely consisting of nonrandomized studies and some randomized

controlled trials consistently demonstrates safety and effectiveness for the use of SRS for the following conditions: acoustic neuromas (vestibular schwannomas); arteriovenous malformations; chordomas and chondrosarcomas of the skull base; craniopharyngiomas; hemangioblastoma; hemangiopericytoma as primary treatment or salvage treatment of recurrence; glomus jugulare and glomus tympanicum tumors; nonresectable, residual, or recurrent meningiomas; metastasis to CNS with adequate performance score and life expectancy meeting policy criteria; pituitary adenomas; primary neoplasms of the central nervous system; trigeminal neuralgia that is refractory to medical management; and uveal melanoma. In addition, clinical practice guidelines recommend the use of SRS for many these indications. Therefore, the use of SRS is considered medically necessary when policy criteria are met for these indications.

The current evidence on stereotactic radiosurgery (SRS) for the treatment of brain metastasis suggests some benefit, including local control and acceptable treatment-related toxicity. However, the studies have significant methodological limitations and the characteristics of patients most likely to derive better health outcomes from SRS compared with other treatment options are not clear. Therefore, except in select patients with good prognosis (Karnofsky score ≥ 70 and life expectancy > 6 months), SRS is considered investigational for the treatment of brain metastases.

The current evidence of SRS for the treatment of refractory symptoms of essential tremor and Parkinson's disease are extremely limited and report conflicting findings. In addition, there is no evidence from these studies comparing SRS with deep brain stimulation or radiofrequency ablation, which are considered the therapies of choice for those with medically refractive disease. Due to inadequate evidence, conclusions cannot be drawn about the safety and effectiveness of SRS for these indications. Therefore, SRS for treatment of Parkinson's disease and essential tremors is considered investigational.

The current evidence base is insufficient for the use of SRS for all other indications. The current evidence doesn't consistently demonstrate improvement in net health outcomes or there is a lack of evidence to demonstrate the effectiveness of SRS. Therefore, the use of SRS for all other indications is considered investigational.

Stereotactic Body Radiotherapy Therapy (i.e., stereotactic ablative radiotherapy [SABR])

Adrenal Metastases

The level of evidence on stereotactic body radiation therapy (SBRT) for adrenal metastases is insufficient to demonstrate the impact of SBRT on patient health outcomes. Therefore, the use of SBRT for adrenal metastases is considered investigational.

Choroidal neovascularization (CNV)

There is insufficient evidence to determine if stereotactic radiotherapy improves net health outcomes in the treatment of choroidal neovascularization (CNV). Therefore, SBRT for CNV is considered investigational.

Chronic Pain

The studies of stereotactic radiosurgery (SRS) and stereotactic body radiation therapy (SBRT) for treatment of chronic pain are extremely limited. This evidence is not sufficient to understand the safety and effectiveness of SBRT for the treatment of chronic pain or to adequately describe the subpopulation

of patients with chronic pain most likely to benefit. Therefore, the use of SRS or SBRT for chronic pain is considered investigational.

Epilepsy

The studies of stereotactic radiosurgery (SRS) for treatment of epilepsy published to date are preliminary in nature, have very small study populations (less than 50 participants), short follow-up times, and/or contain heterogeneous study populations. In addition, the available evidence from patients with epileptic lesions of various sizes and locations is unable to adequately show what factors are associated with favorable outcomes following SRS treatment. Therefore, SRS for epilepsy is considered investigational.

Hepatic Tumors

The current evidence for hepatic tumors is limited. However, there is consensus around the palliative use of stereotactic body radiation therapy (SBRT) for hepatic tumors. Therefore, the use of SBRT for the treatment of hepatic tumors (primary or metastatic) is considered medically necessary when there is minimal extra hepatic disease, the patient has a good performance score, and life expectancy greater than six months. All other indications for the use of SBRT for hepatic tumors is considered investigational.

Hepatocellular Cancer (HCC)

Current studies of stereotactic body radiation therapy (SBRT) for hepatocellular carcinoma (HCC) include heterogeneous patient populations. However, the current evidence consistently shows an improvement in effectiveness outcomes and especially in patients with less than five tumors and less than 6 centimeters in diameter. Therefore, SBRT for the treatment of HCC is considered medically necessary when policy criteria are met. The current evidence base for all other indications is insufficient to establish the effectiveness and safety of SBRT for HCC. Therefore, the use of SBRT for all other indications for HCC is investigational.

Kidney Cancer

The literature on stereotactic body radiation therapy (SBRT) for renal cell carcinoma is limited. The level of evidence is not sufficient to understand the impact of SBRT on renal carcinoma-related health outcomes especially compared to more established treatment modalities. Therefore, the use of SBRT for kidney cancer is considered investigational.

Lung Metastases

Evidence on stereotactic body radiation therapy (SBRT) treatment of lung metastases has methodological limitations; however studies consistently suggest some benefit of SBRT in the treatment of lung metastases (e.g., local control and acceptable treatment-related toxicity) in a select group of patients with a limited number of metastases. Outside this subgroup, there is limited understanding of safety and efficacy of SBRT for lung metastases. Therefore, except in the select group of patients (as defined in the policy criteria), SBRT of lung metastases is considered investigational.

Oligometastases

The evidence related to the use of SBRT for oligometastases to locations not currently addressed as medically necessary in the policy criteria is either insufficient or heterogeneous and limited to small sample sizes. In addition, many of the studies are retrospective, noncomparative analyses. More evidence is needed to establish the role of SBRT for the treatment of oligometastases. Therefore, the use of SBRT for oligometastases not currently addressed as medically necessary in the policy criteria is considered investigational.

Osteosarcoma

The evidence base for the use of stereotactic body radiation therapy (SBRT) for treatment of osteosarcomas is limited. However, there may never be large comparative studies for this population. Current clinical practice guidelines recommend SBRT as a treatment option for osteosarcoma metastatic disease. Therefore, SBRT for the treatment of osteosarcoma metastatic disease may be considered medically necessary. All other indications for the use of SBRT for osteosarcoma is considered investigational.

Pancreatic Cancer

The role of stereotactic body radiation therapy (SBRT) for pancreatic tumors has not been established. At this time, it is not known which patient populations would most likely benefit from this treatment. Although studies have shown promising local control rates, there have been no significant changes in patient survival compared with historical data and some studies have shown unacceptable toxicity and questionable palliative effect. Therefore, the use of SBRT for pancreatic cancer is considered investigational.

Primary Non-Small Cell Lung Cancer

Non-comparative studies have consistently shown that stereotactic body radiation therapy (SBRT) for patients with non-small cell lung cancer (NSCLC), node negative, tumor stage T1a, T1b, T2a, or T2b, have survival rates comparable to patients who have undergone surgical resection. In addition, clinical practice guidelines recommend the use of SBRT for NSCLC. Therefore, SBRT is considered medically necessary for patients with NSCLC, node negative, tumor stage T1a, T1b, T2a, or T2b. SBRT for NSCLC is considered investigational for all other indications.

Prostate Cancer

Stereotactic body radiation therapy (SBRT) has not been compared in scientific studies with established standards of care for radiation of prostate cancer, such as intensity-modulated radiation therapy (IMRT) or 3D CRT. Therefore, the safety and efficacy of SBRT compared with the standards of care is not known. Evidence on SBRT for prostate cancer consists of single-arm studies reporting acute and late toxicity and comparison studies focusing on early PSA outcomes. Although studies have reported promising initial results including low toxicity rates, longer term follow-up is necessary in order to determine long-term toxicity, cancer control, and survival. Therefore, the use of SBRT for prostate cancer is considered investigational.

Spinal and Vertebral Body Tumors (Primary or Metastatic)

The current evidence base suggests that stereotactic body radiation therapy (SBRT) leads to improved net health outcomes in patients with spinal or vertebral body tumors and especially in patients that have received prior radiation therapy. In addition, there is expert clinical consensus on the benefits of SBRT

in this population. Therefore, SBRT is considered medically necessary for the treatment of primary and salvage treatment of local recurrence after previous irradiation when there is clinical documentation of no high grade epidural compression of the spinal cord. The use of SBRT for all other indications for spinal tumors is considered investigational.

Trigeminal Neuralgia

Evidence that shows effectiveness of stereotactic radiosurgery (SRS) for treatment of trigeminal neuralgia comes from the studies of patients with the disease refractory to medical management (i.e., not adequately treated with). This evidence does not allow conclusions concerning safety and efficacy of SRS as a primary treatment for trigeminal neuralgia. Therefore, except in the patients with trigeminal neuralgia refractory to medical management, SRS is considered investigational.

REFERENCES

1. Mohr, JP, Parides, MK, Stapf, C, et al. Medical management with or without interventional therapy for unruptured brain arteriovenous malformations (ARUBA): a multicentre, non-blinded, randomised trial. *Lancet*. 2014 Feb 15;383(9917):614-21. PMID: 24268105
2. Paul, L, Casasco, A, Kusak, ME, Martinez, N, Rey, G, Martinez, R. Results for a Series of 697 AVMs Treated by Gamma Knife: Influence of Angiographic Features on the Obliteration Rate. *Neurosurgery*. 2014 Jul 18. PMID: 25050575
3. Bowden, G, Kano, H, Tonetti, D, Niranjan, A, Flickinger, J, Lunsford, LD. Stereotactic radiosurgery for arteriovenous malformations of the cerebellum. *J Neurosurg*. 2014 Mar;120(3):583-90. PMID: 24160482
4. Fokas, E, Henzel, M, Wittig, A, Grund, S, Engenhart-Cabillic, R. Stereotactic radiosurgery of cerebral arteriovenous malformations: long-term follow-up in 164 patients of a single institution. *Journal of neurology*. 2013 Aug;260(8):2156-62. PMID: 23712798
5. Matsuo, T, Kamada, K, Izumo, T, Hayashi, N, Nagata, I. Linear accelerator-based radiosurgery alone for arteriovenous malformation: more than 12 years of observation. *Int J Radiat Oncol Biol Phys*. 2014 Jul 1;89(3):576-83. PMID: 24803036
6. Potts, MB, Sheth, SA, Louie, J, et al. Stereotactic radiosurgery at a low marginal dose for the treatment of pediatric arteriovenous malformations: obliteration, complications, and functional outcomes. *Journal of neurosurgery Pediatrics*. 2014 Jul;14(1):1-11. PMID: 24766309
7. Kano, H, Kondziolka, D, Flickinger, JC, et al. Stereotactic radiosurgery for arteriovenous malformations, Part 6: multistaged volumetric management of large arteriovenous malformations. *J Neurosurg*. 2012 Jan;116(1):54-65. PMID: 22077447
8. Yen, CP, Schlesinger, D, Sheehan, JP. Gamma Knife(R) radiosurgery for trigeminal neuralgia. *Expert review of medical devices*. 2011 Nov;8(6):709-21. PMID: 22029468
9. Dhople, AA, Adams, JR, Maggio, WW, Naqvi, SA, Regine, WF, Kwok, Y. Long-term outcomes of Gamma Knife radiosurgery for classic trigeminal neuralgia: implications of treatment and critical review of the literature. Clinical article. *J Neurosurg*. 2009 Aug;111(2):351-8. PMID: 19326987
10. Special report: stereotactic radiosurgery for intracranial lesions by gamma beam, linear accelerator, and proton beam methods. *Tecnologica MAP supplement Blue Cross and Blue Shield Association Medical Advisory Panel*. 1999 Jan;26-7. PMID: 10346748
11. Regis, J, Bartolomei, F, Rey, M, Hayashi, M, Chauvel, P, Peragut, JC. Gamma knife surgery for mesial temporal lobe epilepsy. *J Neurosurg*. 2000 Dec;93 Suppl 3:141-6. PMID: 11143232

12. Schrottner, O, Eder, HG, Unger, F, Feichtinger, K, Pendl, G. Radiosurgery in lesional epilepsy: brain tumors. *Stereotact Funct Neurosurg.* 1998 Oct;70 Suppl 1:50-6. PMID: 9782235
13. Whang, CJ, Kwon, Y. Long-term follow-up of stereotactic Gamma Knife radiosurgery in epilepsy. *Stereotact Funct Neurosurg.* 1996;66 Suppl 1:349-56. PMID: 9032879
14. Kondziolka, D, Ong, JG, Lee, JY, Moore, RY, Flickinger, JC, Lunsford, LD. Gamma Knife thalamotomy for essential tremor. *J Neurosurg.* 2008 Jan;108(1):111-7. PMID: 18173319
15. Kooshkabad, A, Lunsford, LD, Tonetti, D, Flickinger, JC, Kondziolka, D. Gamma Knife thalamotomy for tremor in the magnetic resonance imaging era. *J Neurosurg.* 2013 Apr;118(4):713-8. PMID: 23373801
16. Lim, SY, Hodaie, M, Fallis, M, Poon, YY, Mazzella, F, Moro, E. Gamma knife thalamotomy for disabling tremor: a blinded evaluation. *Archives of neurology.* 2010 May;67(5):584-8. PMID: 20457958
17. Ohye, C, Higuchi, Y, Shibasaki, T, et al. Gamma knife thalamotomy for Parkinson disease and essential tremor: a prospective multicenter study. *Neurosurgery.* 2012 Mar;70(3):526-35; discussion 35-6. PMID: 21904267
18. Young, RF, Jacques, S, Mark, R, et al. Gamma knife thalamotomy for treatment of tremor: long-term results. *J Neurosurg.* 2000 Dec;93 Suppl 3:128-35. PMID: 11143229
19. Badakhshi, H, Muellner, S, Wiener, E, Budach, V. Image-guided stereotactic radiotherapy for patients with vestibular schwannoma. A clinical study. *Strahlenther Onkol.* 2014 Jun;190(6):533-7. PMID: 24589920
20. Williams, BJ, Xu, Z, Salvetti, DJ, McNeill, IT, Larnier, J, Sheehan, JP. Gamma Knife surgery for large vestibular schwannomas: a single-center retrospective case-matched comparison assessing the effect of lesion size. *J Neurosurg.* 2013 Aug;119(2):463-71. PMID: 23706053
21. Pollock, BE, Driscoll, CL, Foote, RL, et al. Patient outcomes after vestibular schwannoma management: a prospective comparison of microsurgical resection and stereotactic radiosurgery. *Neurosurgery.* 2006 Jul;59(1):77-85; discussion 77-85. PMID: 16823303
22. Meijer, OW, Vandertop, WP, Baayen, JC, Slotman, BJ. Single-fraction vs. fractionated linac-based stereotactic radiosurgery for vestibular schwannoma: a single-institution study. *Int J Radiat Oncol Biol Phys.* 2003 Aug 1;56(5):1390-6. PMID: 12873685
23. Chung, HT, Ma, R, Toyota, B, Clark, B, Robar, J, McKenzie, M. Audiologic and treatment outcomes after linear accelerator-based stereotactic irradiation for acoustic neuroma. *Int J Radiat Oncol Biol Phys.* 2004 Jul 15;59(4):1116-21. PMID: 15234046
24. Chang, SD, Gibbs, IC, Sakamoto, GT, Lee, E, Oyelese, A, Adler, JR, Jr. Staged stereotactic irradiation for acoustic neuroma. *Neurosurgery.* 2005 Jun;56(6):1254-61; discussion 61-3. PMID: 15918941
25. Hashizume, C, Mori, Y, Kobayashi, T, Shibamoto, Y, Nagai, A, Hayashi, N. Stereotactic radiotherapy using Novalis for craniopharyngioma adjacent to optic pathways. *Journal of neuro-oncology.* 2010 Jun;98(2):239-47. PMID: 20422439
26. Hasegawa, T, Kobayashi, T, Kida, Y. Tolerance of the optic apparatus in single-fraction irradiation using stereotactic radiosurgery: evaluation in 100 patients with craniopharyngioma. *Neurosurgery.* 2010 Apr;66(4):688-94; discussion 94-5. PMID: 20190668
27. Combs, SE, Thilmann, C, Huber, PE, Hoess, A, Debus, J, Schulz-Ertner, D. Achievement of long-term local control in patients with craniopharyngiomas using high precision stereotactic radiotherapy. *Cancer.* 2007 Jun 1;109(11):2308-14. PMID: 17469176
28. Sheehan, JP, Kano, H, Xu, Z, et al. Gamma Knife radiosurgery for facial nerve schwannomas: a multicenter study. *J Neurosurg.* 2015 Aug;123(2):387-94. PMID: 25955875
29. Ivan, ME, Sughrue, ME, Clark, AJ, et al. A meta-analysis of tumor control rates and treatment-related morbidity for patients with glomus jugulare tumors. *J Neurosurg.* 2011 May;114(5):1299-305. PMID: 21029039

30. Kano, H, Shuto, T, Iwai, Y, et al. Stereotactic radiosurgery for intracranial hemangioblastomas: a retrospective international outcome study. *J Neurosurg.* 2015 Jun;122(6):1469-78. PMID: 25816088
31. Sayer, FT, Nguyen, J, Starke, RM, Yen, CP, Sheehan, JP. Gamma knife radiosurgery for intracranial hemangioblastomas--outcome at 3 years. United States, 2011. p. 99-105; discussion 45-8.
32. Puataweepong, P, Dhanachai, M, Hansasuta, A, et al. The clinical outcome of intracranial hemangioblastomas treated with linac-based stereotactic radiosurgery and radiotherapy. England, 2014. p. 761-8.
33. Hanakita, S, Koga, T, Shin, M, et al. The long-term outcomes of radiosurgery for intracranial hemangioblastomas. England, 2014. p. 429-33.
34. Asthagiri, AR, Mehta, GU, Zach, L, et al. Prospective evaluation of radiosurgery for hemangioblastomas in von Hippel-Lindau disease. United States, 2010. p. 80-6.
35. Moss, JM, Choi, CY, Adler, JR, Jr., Soltys, SG, Gibbs, IC, Chang, SD. Stereotactic radiosurgical treatment of cranial and spinal hemangioblastomas. United States, 2009. p. 79-85; discussion
36. Kano, H, Niranjana, A, Mongia, S, Kondziolka, D, Flickinger, JC, Lunsford, LD. The role of stereotactic radiosurgery for intracranial hemangioblastomas. United States, 2008. p. 443-50; discussion 50-1.
37. Patrice, SJ, Sneed, PK, Flickinger, JC, et al. Radiosurgery for hemangioblastoma: results of a multiinstitutional experience. United States, 1996. p. 493-9.
38. Chang, SD, Meisel, JA, Hancock, SL, Martin, DP, McManus, M, Adler, JR, Jr. Treatment of hemangioblastomas in von Hippel-Lindau disease with linear accelerator-based radiosurgery. *Neurosurgery.* 1998 Jul;43(1):28-34; discussion -5. PMID: 9657185
39. Jawahar, A, Kondziolka, D, Garces, YI, Flickinger, JC, Pollock, BE, Lunsford, LD. Stereotactic radiosurgery for hemangioblastomas of the brain. *Acta neurochirurgica.* 2000;142(6):641-4; discussion 4-5. PMID: 10949438
40. Tago, M, Terahara, A, Shin, M, et al. Gamma knife surgery for hemangioblastomas. *J Neurosurg.* 2005 Jan;102 Suppl:171-4. PMID: 15662804
41. Chen, Y, Li, ZF, Zhang, FX, et al. Gamma knife surgery for patients with volumetric classification of nonfunctioning pituitary adenomas: a systematic review and meta-analysis. *European journal of endocrinology / European Federation of Endocrine Societies.* 2013 Oct;169(4):487-95. PMID: 23904281
42. Lee, CC, Kano, H, Yang, HC, et al. Initial Gamma Knife radiosurgery for nonfunctioning pituitary adenomas. *J Neurosurg.* 2014 Mar;120(3):647-54. PMID: 24405068
43. Sheehan, JP, Starke, RM, Mathieu, D, et al. Gamma Knife radiosurgery for the management of nonfunctioning pituitary adenomas: a multicenter study. *J Neurosurg.* 2013 Aug;119(2):446-56. PMID: 23621595
44. El-Shehaby, AM, Reda, WA, Abdel Karim, KM, Emad Eldin, RM, Esene, IN. Gamma Knife radiosurgery for low-grade tectal gliomas. *Acta neurochirurgica.* 2015 Feb;157(2):247-56. PMID: 25510647
45. Clark, GM, McDonald, AM, Nabors, LB, et al. Hypofractionated stereotactic radiosurgery with concurrent bevacizumab for recurrent malignant gliomas: the University of Alabama at Birmingham experience. 2014. p. 172-7.
46. Cabrera, AR, Cuneo, KC, Desjardins, A, et al. Concurrent stereotactic radiosurgery and bevacizumab in recurrent malignant gliomas: a prospective trial. *Int J Radiat Oncol Biol Phys.* 2013 Aug 1;86(5):873-9. PMID: 23725997
47. Cuneo, KC, Vredenburgh, JJ, Sampson, JH, et al. Safety and efficacy of stereotactic radiosurgery and adjuvant bevacizumab in patients with recurrent malignant gliomas. United States, 2012. p. 2018-24.

48. Doodoo, E, Huffmann, B, Peredo, I, et al. Increased survival using delayed gamma knife radiosurgery for recurrent high-grade glioma: a feasibility study. *World neurosurgery*. 2014 Nov;82(5):e623-32. PMID: 24930898
49. Roos, D. What is the randomised evidence for surgery and stereotactic radiosurgery for patients with solitary (or few) brain metastases? *International journal of evidence-based healthcare*. 2011 Mar;9(1):61-6. PMID: 21332664
50. Park, HS, Chiang, VL, Knisely, JP, Raldow, AC, Yu, JB. Stereotactic radiosurgery with or without whole-brain radiotherapy for brain metastases: an update. *Expert review of anticancer therapy*. 2011 Nov;11(11):1731-8. PMID: 22050022
51. Patil, CG, Pricola, K, Garg, SK, Bryant, A, Black, KL. Whole brain radiation therapy (WBRT) alone versus WBRT and radiosurgery for the treatment of brain metastases. *The Cochrane database of systematic reviews*. 2010(6):CD006121. PMID: 20556764
52. Chang, DT, Schellenberg, D, Shen, J, et al. Stereotactic radiotherapy for unresectable adenocarcinoma of the pancreas. *Cancer*. 2009 Feb 1;115(3):665-72. PMID: 19117351
53. Kondziolka, D, Patel, A, Lunsford, LD, Kassam, A, Flickinger, JC. Stereotactic radiosurgery plus whole brain radiotherapy versus radiotherapy alone for patients with multiple brain metastases. *Int J Radiat Oncol Biol Phys*. 1999 Sep 1;45(2):427-34. PMID: 10487566
54. Weltman, E, Salvajoli, JV, Brandt, RA, et al. Radiosurgery for brain metastases: a score index for predicting prognosis. *Int J Radiat Oncol Biol Phys*. 2000 Mar 15;46(5):1155-61. PMID: 10725626
55. Yu, C, Chen, JC, Apuzzo, ML, et al. Metastatic melanoma to the brain: prognostic factors after gamma knife radiosurgery. *Int J Radiat Oncol Biol Phys*. 2002 Apr 1;52(5):1277-87. PMID: 11955740
56. Aoyama, H, Shirato, H, Tago, M, et al. Stereotactic radiosurgery plus whole-brain radiation therapy vs stereotactic radiosurgery alone for treatment of brain metastases: a randomized controlled trial. *JAMA*. 2006 Jun 7;295(21):2483-91. PMID: 16757720
57. Raizer, J. Radiosurgery and whole-brain radiation therapy for brain metastases: either or both as the optimal treatment. *JAMA*. 2006 Jun 7;295(21):2535-6. PMID: 16757726
58. Tian, LJ, Zhuang, HQ, Yuan, ZY. A comparison between cyberknife and neurosurgery in solitary brain metastases from non-small cell lung cancer. *Clinical neurology and neurosurgery*. 2013 Oct;115(10):2009-14. PMID: 23850045
59. Yamamoto, M, Serizawa, T, Shuto, T, et al. Stereotactic radiosurgery for patients with multiple brain metastases (JLGK0901): a multi-institutional prospective observational study. *Lancet Oncol*. 2014 Apr;15(4):387-95. PMID: 24621620
60. Rava, P, Leonard, K, Sioshansi, S, et al. Survival among patients with 10 or more brain metastases treated with stereotactic radiosurgery. *J Neurosurg*. 2013 Aug;119(2):457-62. PMID: 23662828
61. Raldow, AC, Chiang, VL, Knisely, JP, Yu, JB. Survival and intracranial control of patients with 5 or more brain metastases treated with gamma knife stereotactic radiosurgery. *Am J Clin Oncol*. 2013 Oct;36(5):486-90. PMID: 22706180
62. Yomo, S, Hayashi, M. Upfront stereotactic radiosurgery in patients with brain metastases from small cell lung cancer: retrospective analysis of 41 patients. *Radiat Oncol*. 2014;9(1):152. PMID: 25005424
63. Zehetmayer, M. Stereotactic photon beam irradiation of uveal melanoma. *Developments in ophthalmology*. 2012;49:58-65. PMID: 22042013
64. Dunavoelgyi, R, Dieckmann, K, Gleiss, A, et al. Local tumor control, visual acuity, and survival after hypofractionated stereotactic photon radiotherapy of choroidal melanoma in 212 patients treated between 1997 and 2007. *Int J Radiat Oncol Biol Phys*. 2011 Sep 1;81(1):199-205. PMID: 20675066

65. Wackernagel, W, Holl, E, Tarmann, L, et al. Local tumour control and eye preservation after gamma-knife radiosurgery of choroidal melanomas. *The British journal of ophthalmology*. 2014 Feb;98(2):218-23. PMID: 24169651
66. Furdova, A, Sramka, M, Chorvath, M, et al. Stereotactic radiosurgery in intraocular malignant melanoma--retrospective study. *Neuro endocrinology letters*. 2014;35(1):28-36. PMID: 24625918
67. Sarici, AM, Pazarli, H. Gamma-knife-based stereotactic radiosurgery for medium- and large-sized posterior uveal melanoma. *Graefe's archive for clinical and experimental ophthalmology = Albrecht von Graefes Archiv fur klinische und experimentelle Ophthalmologie*. 2013 Jan;251(1):285-94. PMID: 22944897
68. Muller, K, Naus, N, Nowak, PJ, et al. Fractionated stereotactic radiotherapy for uveal melanoma, late clinical results. *Radiother Oncol*. 2012 Feb;102(2):219-24. PMID: 21864922
69. Furdova, A, Slezak, P, Chorvath, M, Waczulikova, I, Sramka, M, Kralik, G. No differences in outcome between radical surgical treatment (enucleation) and stereotactic radiosurgery in patients with posterior uveal melanoma. *Neoplasma*. 2010;57(4):377-81. PMID: 20429631
70. Gerszten, PC, Ozhasoglu, C, Burton, SA, et al. CyberKnife frameless stereotactic radiosurgery for spinal lesions: clinical experience in 125 cases. *Neurosurgery*. 2004 Jul;55(1):89-98; discussion 9. PMID: 15214977
71. Degen, JW, Gagnon, GJ, Voyadzis, JM, et al. CyberKnife stereotactic radiosurgical treatment of spinal tumors for pain control and quality of life. *J Neurosurg Spine*. 2005 May;2(5):540-9. PMID: 15945428
72. Gerszten, PC, Burton, SA, Ozhasoglu, C, Welch, WC. Radiosurgery for spinal metastases: clinical experience in 500 cases from a single institution. *Spine (Phila Pa 1976)*. 2007 Jan 15;32(2):193-9. PMID: 17224814
73. Chang, EL, Shiu, AS, Mendel, E, et al. Phase I/II study of stereotactic body radiotherapy for spinal metastasis and its pattern of failure. *J Neurosurg Spine*. 2007 Aug;7(2):151-60. PMID: 17688054
74. Sahgal, A, Atenafu, EG, Chao, S, et al. Vertebral compression fracture after spine stereotactic body radiotherapy: a multi-institutional analysis with a focus on radiation dose and the spinal instability neoplastic score. *J Clin Oncol*. 2013 Sep 20;31(27):3426-31. PMID: 23960179
75. Zheng, X, Schipper, M, Kidwell, K, et al. Survival Outcome After Stereotactic Body Radiation Therapy and Surgery for Stage I Non-Small Cell Lung Cancer: A Meta-Analysis. *International journal of radiation oncology, biology, physics*. 2014 Jul 19. PMID: 25052562
76. Nguyen, NP, Garland, L, Welsh, J, Hamilton, R, Cohen, D, Vinh-Hung, V. Can stereotactic fractionated radiation therapy become the standard of care for early stage non-small cell lung carcinoma. *Cancer Treat Rev*. 2008 Dec;34(8):719-27. PMID: 18657910
77. Koto, M, Takai, Y, Ogawa, Y, et al. A phase II study on stereotactic body radiotherapy for stage I non-small cell lung cancer. *Radiother Oncol*. 2007 Dec;85(3):429-34. PMID: 18022720
78. Kupelian, PA, Komaki, R, Allen, P. Prognostic factors in the treatment of node-negative nonsmall cell lung carcinoma with radiotherapy alone. *Int J Radiat Oncol Biol Phys*. 1996 Oct 1;36(3):607-13. PMID: 8948345
79. Crabtree, TD, Puri, V, Robinson, C, et al. Analysis of first recurrence and survival in patients with stage I non-small cell lung cancer treated with surgical resection or stereotactic radiation therapy. *The Journal of thoracic and cardiovascular surgery*. 2014 Apr;147(4):1183-91; discussion 91-2. PMID: 24507980
80. Jeppesen, SS, Schytte, T, Jensen, HR, Brink, C, Hansen, O. Stereotactic body radiation therapy versus conventional radiation therapy in patients with early stage non-small cell lung cancer: an updated retrospective study on local failure and survival rates. *Acta Oncol*. 2013 Oct;52(7):1552-8. PMID: 23902274

81. Port, JL, Parashar, B, Osakwe, N, et al. A Propensity-Matched Analysis of Wedge Resection and Stereotactic Body Radiotherapy for Early Stage Lung Cancer. *The Annals of thoracic surgery*. 2014 Jul 29. PMID: 25085557
82. Varlotto, J, Fakiris, A, Flickinger, J, et al. Matched-pair and propensity score comparisons of outcomes of patients with clinical stage I non-small cell lung cancer treated with resection or stereotactic radiosurgery. *Cancer*. 2013 Aug 1;119(15):2683-91. PMID: 23605504
83. Timmerman, RD, Park, C, Kavanagh, BD. The North American experience with stereotactic body radiation therapy in non-small cell lung cancer. *J Thorac Oncol*. 2007 Jul;2(7 Suppl 3):S101-12. PMID: 17603304
84. Hof, H, Muentert, M, Oetzel, D, Hoess, A, Debus, J, Herfarth, K. Stereotactic single-dose radiotherapy (radiosurgery) of early stage nonsmall-cell lung cancer (NSCLC). *Cancer*. 2007 Jul 1;110(1):148-55. PMID: 17516437
85. Allibhai, Z, Taremi, M, Bezjak, A, et al. The impact of tumor size on outcomes after stereotactic body radiation therapy for medically inoperable early-stage non-small cell lung cancer. *Int J Radiat Oncol Biol Phys*. 2013 Dec 1;87(5):1064-70. PMID: 24210082
86. Harkenrider, MM, Bertke, MH, Dunlap, NE. Stereotactic Body Radiation Therapy for Unbiopsied Early-stage Lung Cancer: A Multi-Institutional Analysis. *Am J Clin Oncol*. 2014 Aug;37(4):337-42. PMID: 23660597
87. Meng, MB, Cui, YL, Lu, Y, et al. Transcatheter arterial chemoembolization in combination with radiotherapy for unresectable hepatocellular carcinoma: a systematic review and meta-analysis. *Radiother Oncol*. 2009 Aug;92(2):184-94. PMID: 19042048
88. Tao, C, Yang, LX. Improved radiotherapy for primary and secondary liver cancer: stereotactic body radiation therapy. *Anticancer research*. 2012 Feb;32(2):649-55. PMID: 22287758
89. Scorsetti, M, Comito, T, Cozzi, L, et al. The challenge of inoperable hepatocellular carcinoma (HCC): results of a single-institutional experience on stereotactic body radiation therapy (SBRT). *Journal of cancer research and clinical oncology*. 2015 Jul;141(7):1301-9. PMID: 25644863
90. Bujold, A, Massey, CA, Kim, JJ, et al. Sequential phase I and II trials of stereotactic body radiotherapy for locally advanced hepatocellular carcinoma. *J Clin Oncol*. 2013 May 1;31(13):1631-9. PMID: 23547075
91. Andolino, DL, Johnson, CS, Maluccio, M, et al. Stereotactic body radiotherapy for primary hepatocellular carcinoma. *Int J Radiat Oncol Biol Phys*. 2011 Nov 15;81(4):e447-53. PMID: 21645977
92. Ibarra, RA, Rojas, D, Snyder, L, et al. Multicenter results of stereotactic body radiotherapy (SBRT) for non-resectable primary liver tumors. *Acta Oncol*. 2012 May;51(5):575-83. PMID: 22263926
93. Price, TR, Perkins, SM, Sandrasegaran, K, et al. Evaluation of response after stereotactic body radiotherapy for hepatocellular carcinoma. *Cancer*. 2012 Jun 15;118(12):3191-8. PMID: 22025126
94. Louis, C, Dewas, S, Mirabel, X, et al. Stereotactic radiotherapy of hepatocellular carcinoma: preliminary results. *Technology in cancer research & treatment*. 2010 Oct;9(5):479-87. PMID: 20815419
95. Kwon, JH, Bae, SH, Kim, JY, et al. Long-term effect of stereotactic body radiation therapy for primary hepatocellular carcinoma ineligible for local ablation therapy or surgical resection. Stereotactic radiotherapy for liver cancer. *BMC cancer*. 2010;10:475. PMID: 20813065
96. Yoon, SM, Lim, YS, Park, MJ, et al. Stereotactic body radiation therapy as an alternative treatment for small hepatocellular carcinoma. *PloS one*. 2013;8(11):e79854. PMID: 24255719
97. Jung, J, Yoon, SM, Kim, SY, et al. Radiation-induced liver disease after stereotactic body radiotherapy for small hepatocellular carcinoma: clinical and dose-volumetric parameters. *Radiat Oncol*. 2013;8:249. PMID: 24160910

98. Katz, A, Ferrer, M, Suarez, JF. Comparison of quality of life after stereotactic body radiotherapy and surgery for early-stage prostate cancer. *Radiat Oncol*. 2012;7:194. PMID: 23164305
99. Yu, JB, Cramer, LD, Herrin, J, Soulos, PR, Potosky, AL, Gross, CP. Stereotactic body radiation therapy versus intensity-modulated radiation therapy for prostate cancer: comparison of toxicity. *J Clin Oncol*. 2014 Apr 20;32(12):1195-201. PMID: 24616315
100. McBride, SM, Wong, DS, Dombrowski, JJ, et al. Hypofractionated stereotactic body radiotherapy in low-risk prostate adenocarcinoma: Preliminary results of a multi-institutional phase 1 feasibility trial. *Cancer*. 2011 Dec 13;118(15):3681-90. PMID: 22170628
101. Boike, TP, Lotan, Y, Cho, LC, et al. Phase I dose-escalation study of stereotactic body radiation therapy for low- and intermediate-risk prostate cancer. *J Clin Oncol*. 2011 May 20;29(15):2020-6. PMID: 21464418
102. Freeman, DE, King, CR. Stereotactic body radiotherapy for low-risk prostate cancer: five-year outcomes. *Radiat Oncol*. 2011;6:3. PMID: 21219625
103. Jabbari, S, Weinberg, VK, Kaprealian, T, et al. Stereotactic body radiotherapy as monotherapy or post-external beam radiotherapy boost for prostate cancer: technique, early toxicity, and PSA response. *Int J Radiat Oncol Biol Phys*. 2012 Jan 1;82(1):228-34. PMID: 21183287
104. King, CR, Brooks, JD, Gill, H, Presti, JC, Jr. Long-term outcomes from a prospective trial of stereotactic body radiotherapy for low-risk prostate cancer. *Int J Radiat Oncol Biol Phys*. 2012 Feb 1;82(2):877-82. PMID: 21300474
105. Wiegner, EA, King, CR. Sexual function after stereotactic body radiotherapy for prostate cancer: results of a prospective clinical trial. *Int J Radiat Oncol Biol Phys*. 2010 Oct 1;78(2):442-8. PMID: 20137864
106. Katz, AJ, Santoro, M, Ashley, R, Diblasio, F, Witten, M. Stereotactic body radiotherapy for organ-confined prostate cancer. *BMC urology*. 2010;10:1. PMID: 20122161
107. Katz, AJ, Santoro, M, Diblasio, F, Ashley, R. Stereotactic body radiotherapy for localized prostate cancer: disease control and quality of life at 6 years. *Radiat Oncol*. 2013;8:118. PMID: 23668632
108. Bolzicco, G, Favretto, MS, Satariano, N, Scremin, E, Tambone, C, Tasca, A. A single-center study of 100 consecutive patients with localized prostate cancer treated with stereotactic body radiotherapy. *BMC urology*. 2013;13:49. PMID: 24134138
109. Chen, LN, Suy, S, Wang, H, et al. Patient-reported urinary incontinence following stereotactic body radiation therapy (SBRT) for clinically localized prostate cancer. *Radiat Oncol*. 2014;9:148. PMID: 24966110
110. Kim, DW, Cho, LC, Straka, C, et al. Predictors of rectal tolerance observed in a dose-escalated phase 1-2 trial of stereotactic body radiation therapy for prostate cancer. *Int J Radiat Oncol Biol Phys*. 2014 Jul 1;89(3):509-17. PMID: 24929162
111. King, CR, Collins, S, Fuller, D, et al. Health-related quality of life after stereotactic body radiation therapy for localized prostate cancer: results from a multi-institutional consortium of prospective trials. *Int J Radiat Oncol Biol Phys*. 2013 Dec 1;87(5):939-45. PMID: 24119836
112. Goyal, K, Einstein, D, Ibarra, RA, et al. Stereotactic body radiation therapy for nonresectable tumors of the pancreas. *The Journal of surgical research*. 2012 May 15;174(2):319-25. PMID: 21937061
113. Rwigema, JC, Parikh, SD, Heron, DE, et al. Stereotactic body radiotherapy in the treatment of advanced adenocarcinoma of the pancreas. *Am J Clin Oncol*. 2011 Feb;34(1):63-9. PMID: 20308870
114. Siva, S, Pham, D, Gill, S, Corcoran, NM, Foroudi, F. A systematic review of stereotactic radiotherapy ablation for primary renal cell carcinoma. *BJU international*. 2012 Dec;110(11 Pt B):E737-43. PMID: 23107102

115. Beitler, JJ, Makara, D, Silverman, P, Lederman, G. Definitive, high-dose-per-fraction, conformal, stereotactic external radiation for renal cell carcinoma. *Am J Clin Oncol*. 2004 Dec;27(6):646-8. PMID: 15577450
116. Ranck, MC, Golden, DW, Corbin, KS, et al. Stereotactic body radiotherapy for the treatment of oligometastatic renal cell carcinoma. *Am J Clin Oncol*. 2013 Dec;36(6):589-95. PMID: 22868242
117. Alongi, F, Arcangeli, S, Filippi, AR, Ricardi, U, Scorsetti, M. Review and uses of stereotactic body radiation therapy for oligometastases. *The oncologist*. 2012;17(8):1100-7. PMID: 22723509
118. Corbin, KS, Hellman, S, Weichselbaum, RR. Extracranial oligometastases: a subset of metastases curable with stereotactic radiotherapy. *J Clin Oncol*. 2013 Apr 10;31(11):1384-90. PMID: 23460715
119. Tree, AC, Khoo, VS, Eeles, RA, et al. Stereotactic body radiotherapy for oligometastases. *Lancet Oncol*. 2013 Jan;14(1):e28-37. PMID: 23276369
120. Milano, MT, Katz, AW, Zhang, H, Okunieff, P. Oligometastases treated with stereotactic body radiotherapy: long-term follow-up of prospective study. *Int J Radiat Oncol Biol Phys*. 2012 Jul 1;83(3):878-86. PMID: 22172903
121. Scorsetti, M, Clerici, E, Navarria, P, et al. The role of stereotactic body radiation therapy (SBRT) in the treatment of oligometastatic disease in the elderly. *The British journal of radiology*. 2015 Sep;88(1053):20150111. PMID: 26183933
122. Long-term results of lung metastasectomy: prognostic analyses based on 5206 cases. The International Registry of Lung Metastases. *The Journal of thoracic and cardiovascular surgery*. 1997 Jan;113(1):37-49. PMID: 9011700
123. Siva, S, MacManus, M, Ball, D. Stereotactic radiotherapy for pulmonary oligometastases: a systematic review. *J Thorac Oncol*. 2010 Jul;5(7):1091-9. PMID: 20479693
124. Norihisa, Y, Nagata, Y, Takayama, K, et al. Stereotactic body radiotherapy for oligometastatic lung tumors. *Int J Radiat Oncol Biol Phys*. 2008 Oct 1;72(2):398-403. PMID: 18374506
125. Rusthoven, KE, Kavanagh, BD, Cardenes, H, et al. Multi-institutional phase I/II trial of stereotactic body radiation therapy for liver metastases. *J Clin Oncol*. 2009 Apr 1;27(10):1572-8. PMID: 19255321
126. Osti, MF, Carnevale, A, Valeriani, M, et al. Clinical outcomes of single dose stereotactic radiotherapy for lung metastases. *Clin Lung Cancer*. 2013 Nov;14(6):699-703. PMID: 23886798
127. Chang, DT, Swaminath, A, Kozak, M, et al. Stereotactic body radiotherapy for colorectal liver metastases: A pooled analysis. *Cancer*. 2011 Sep 1;117(17):4060-9. PMID: 21432842
128. Scorsetti, M, Alongi, F, Filippi, AR, et al. Long-term local control achieved after hypofractionated stereotactic body radiotherapy for adrenal gland metastases: a retrospective analysis of 34 patients. *Acta Oncol*. 2012 May;51(5):618-23. PMID: 22263925
129. Holy, R, Piroth, M, Pinkawa, M, Eble, MJ. Stereotactic body radiation therapy (SBRT) for treatment of adrenal gland metastases from non-small cell lung cancer. *Strahlenther Onkol*. 2011 Apr;187(4):245-51. PMID: 21424513
130. Casamassima, F, Livi, L, Masciullo, S, et al. Stereotactic radiotherapy for adrenal gland metastases: university of Florence experience. *Int J Radiat Oncol Biol Phys*. 2012 Feb 1;82(2):919-23. PMID: 21300473
131. Chawla, S, Chen, Y, Katz, AW, et al. Stereotactic body radiotherapy for treatment of adrenal metastases. *Int J Radiat Oncol Biol Phys*. 2009 Sep 1;75(1):71-5. PMID: 19250766
132. Ahmed, KA, Barney, BM, Macdonald, OK, et al. Stereotactic body radiotherapy in the treatment of adrenal metastases. *Am J Clin Oncol*. 2013 Oct;36(5):509-13. PMID: 22781389

133. Napieralska, A, Miszczyk, L, Tukiendorf, A, Stapor-Fudzinska, M. The results of treatment of prostate cancer bone metastases after CyberKnife radiosurgery. *Ortopedia, traumatologia, rehabilitacja*. 2014 Jul 3;16(3):339-49. PMID: 25058109
134. National Comprehensive Cancer Network (NCCN). NCCN Guidelines for Treatment of Cancer by Site. [cited 01/15/2016]; Available from: http://www.nccn.org/professionals/physician_gls/f_guidelines.asp#site
135. Howington, JA, Blum, MG, Chang, AC, Balekian, AA, Murthy, SC. Treatment of stage I and II non-small cell lung cancer: Diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2013 May;143(5 Suppl):e278S-313S. PMID: 23649443
136. Donington, J, Ferguson, M, Mazzone, P, et al. American College of Chest Physicians and Society of Thoracic Surgeons consensus statement for evaluation and management for high-risk patients with stage I non-small cell lung cancer. United States, 2012. p. 1620-35.
137. Simoff, MJ, Lally, B, Slade, MG, et al. Symptom management in patients with lung cancer: Diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2013;143(5 Suppl):e455S-97S. PMID: 23649452
138. Tsao, MN, Rades, D, Wirth, A, et al. Radiotherapeutic and surgical management for newly diagnosed brain metastasis(es): An American Society for Radiation Oncology evidence-based guideline. *Practical radiation oncology*. 2012 Jul-Sep;2(3):210-25. PMID: 25925626
139. Lutz, S, Berk, L, Chang, E, et al. Palliative radiotherapy for bone metastases: an ASTRO evidence-based guideline. United States, 2011. p. 965-76.
140. Ramakrishna, N, Temin, S, Chandarlapaty, S, et al. Recommendations on disease management for patients with advanced human epidermal growth factor receptor 2-positive breast cancer and brain metastases: American Society of Clinical Oncology clinical practice guideline. *J Clin Oncol* 2014 [cited 19]; 2100-8]. Available from:

CROSS REFERENCES

[Charged Particle \(Proton or Helium Ion\) Radiation Therapy](#), Medicine, Policy No. 49

[Intraocular Radiation Therapy for Age-Related Macular Degeneration](#), Medicine, Policy No. 134

[Intensity Modulated Radiation Therapy \(IMRT\) of the Thorax](#), Medicine, Policy No. 136

[Intensity Modulated Radiation Therapy \(IMRT\) of the Prostate](#), Medicine, Policy No. 137

[Intensity Modulated Radiation Therapy \(IMRT\) for Head and Neck Cancers and Thyroid Cancer](#), Medicine, Policy No. 138

[Intensity Modulated Radiation Therapy \(IMRT\) of the Abdomen and Pelvis](#), Medicine, Policy No. 139

[Intensity Modulated Radiation Therapy \(IMRT\): Central Nervous System \(CNS\) and Vertebral Tumors](#), Medicine, Policy No. 147

[Electromagnetic Navigation Bronchoscopy](#), Surgery, Policy No. 179

CODES	NUMBER	DESCRIPTION
Coding for stereotactic radiosurgery typically consists of a series of CPT codes describing the individual steps required; medical radiation physics, clinical treatment planning, attachment of stereotactic head frame, treatment delivery and clinical treatment management.		
Treatment delivery:		
The codes used for treatment delivery will depend on the energy source used, typically either photons or protons. For photons (i.e. with a Gamma knife or LINAC device (including Cyberknife®) nonspecific radiation therapy treatment delivery CPT codes may be used based on the voltage of the energy source (i.e. codes 77402-77416). When proton beam therapy is used CPT codes 77520 thru 77525 are available.		
CPT	32701	Thoracic target(s) delineation for stereotactic body radiation therapy (SRS/SBRT), (photon or particle beam), entire course of treatment
	77371	Radiation therapy delivery, stereotactic radiosurgery (SRS), complete course of treatment of cranial lesion(s) consisting of 1 session; multi-source Cobalt 60 based
	77372	Radiation therapy delivery, stereotactic radiosurgery (SRS), complete course of treatment of cranial lesion(s) consisting of 1 session; linear accelerator based
	77373	Stereotactic body radiation therapy, treatment delivery, per fraction to 1 or more lesions, including image guidance, entire course not to exceed 5 fraction
	77435	Stereotactic body radiation therapy, treatment management, per treatment course, to 1 or more lesions, including image guidance, entire course not to exceed 5 fractions
Note: Codes for treatment delivery primarily reflects the cost related to the energy source used, and not physician work.		
Clinical treatment management:		
CPT	77432	Stereotactic radiation treatment management of cerebral lesion(s) (complete course of treatment consisting of one session.)
	61796	Stereotactic radiosurgery (particle beam, gamma ray, or linear accelerator); 1 simple cranial lesion
	61797	Stereotactic radiosurgery (particle beam, gamma ray, or linear accelerator); each additional cranial lesion, simple (List separately in addition to code for primary procedure)
	61798	Stereotactic radiosurgery (particle beam, gamma ray, or linear accelerator);

CODES	NUMBER	DESCRIPTION
		1 complex cranial lesion
	61799	Stereotactic radiosurgery (particle beam, gamma ray, or linear accelerator); each additional cranial lesion, complex (List separately in addition to code for primary procedure)
	61800	Application of stereotactic headframe for stereotactic radiosurgery (List separately in addition to code for primary procedure)
	63620	Stereotactic radiosurgery (particle beam, gamma ray, or linear accelerator); 1 spinal lesion
	63621	Stereotactic radiosurgery (particle beam, gamma ray, or linear accelerator); each additional spinal lesion (List separately in addition to code for primary procedure)
HCPCS	G0339	Image guided robotic linear accelerator-based stereotactic radiosurgery, complete course of therapy in one session, or first session of fractionated treatment.
	G0340	Image guided robotic linear accelerator-based stereotactic radiosurgery, delivery including collimator changes and custom plugging, fractionated treatment, all lesions, per session, second through fifth sessions, maximum five sessions per course of treatment